

# Future India 2050

Where the trend lines point — using only the projected series the primary sources themselves publish.

|                      |                         |                             |                                    |
|----------------------|-------------------------|-----------------------------|------------------------------------|
| <b>5</b><br>CHAPTERS | <b>45</b><br>INDICATORS | <b>9</b><br>WITH STATE DATA | <b>1951–2036</b><br>PERIOD COVERED |
|----------------------|-------------------------|-----------------------------|------------------------------------|

Generated 21 Aug 2026 · assembled from 40 sources

This PDF is a snapshot. Indicators are revised as their sources publish;  
indiainsightlab.com always carries the current figures.

## EXECUTIVE SUMMARY

This report assembles 45 indicators across 5 chapters, drawn from 40 sources and covering 1951–2036.

Of those indicators, 9 carry state-level detail and can be mapped across India’s states. The remaining 36 are published at national level only by their source, and are shown here as national series without a sub-national breakdown.

**Coverage limits.** This report contains no district- or city-level figures. Comparable district data is not currently integrated into India Insight Lab, and no value here has been estimated, interpolated or carried down from a higher geography to fill that gap. Where a number is absent, it is absent.

## KEY FINDINGS

Each finding below is computed directly from the state values in this report — the highest and lowest recorded state for that indicator. None are authored interpretations.

### 1 State-wise merchandise exports

Ranges from 90\$bn in Maharashtra to 22\$bn in Uttar Pradesh — a spread of 68\$bn across the 6 states reported.

Trade Intelligence and Analytics portal, Department of Commerce

2 **Forest cover by state**

Ranges from 85.4% in Mizoram to 3.6% in Haryana — a spread of 81.8% across the 8 states reported.

India State of Forest Report, Forest Survey of India

3 **Slum population share by state**

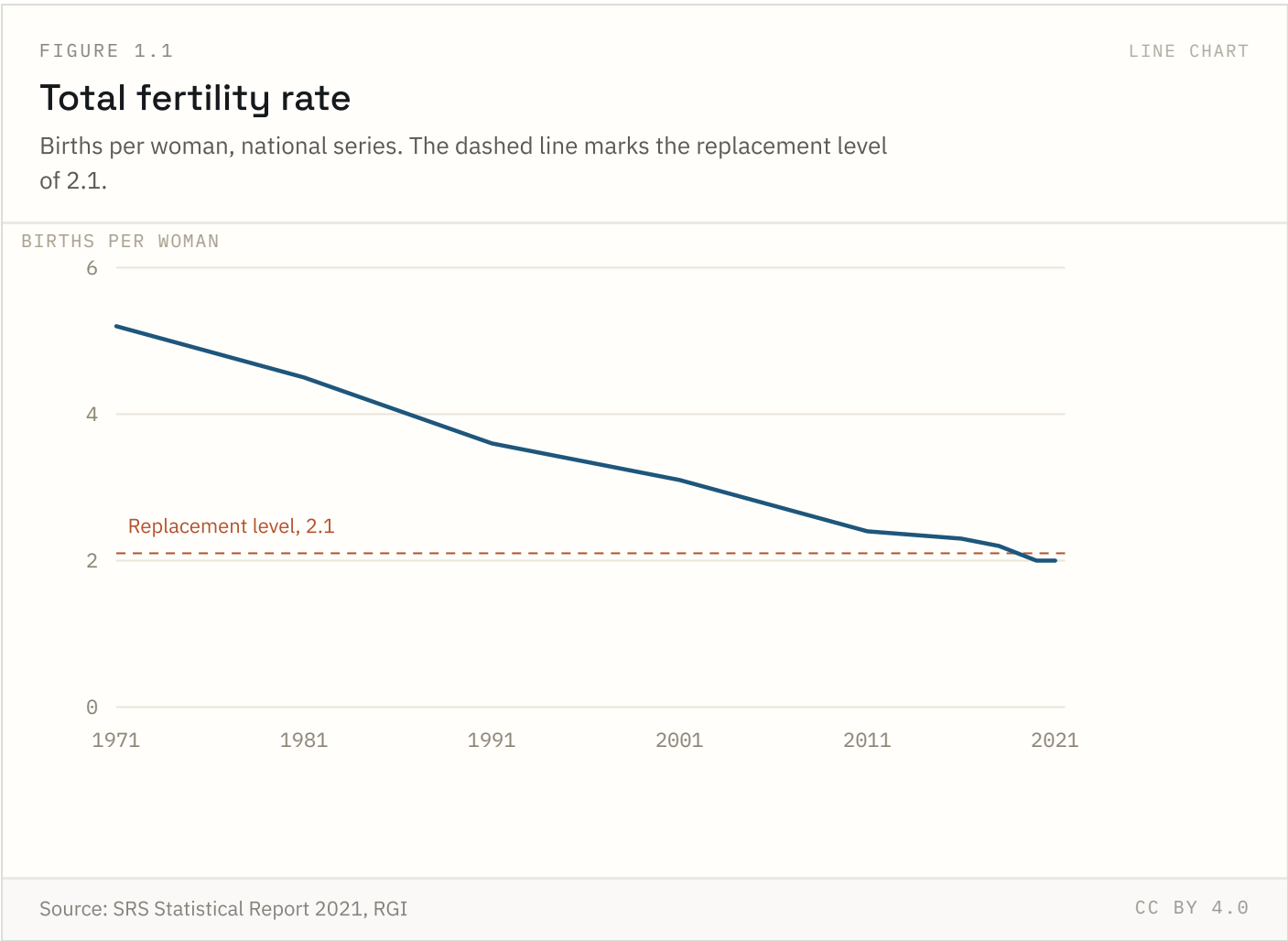
Ranges from 24.0% in Andhra Pradesh to 4.9% in Kerala — a spread of 19.1% across the 15 states reported.

Census of India 2011, Primary Census Abstract for Slum

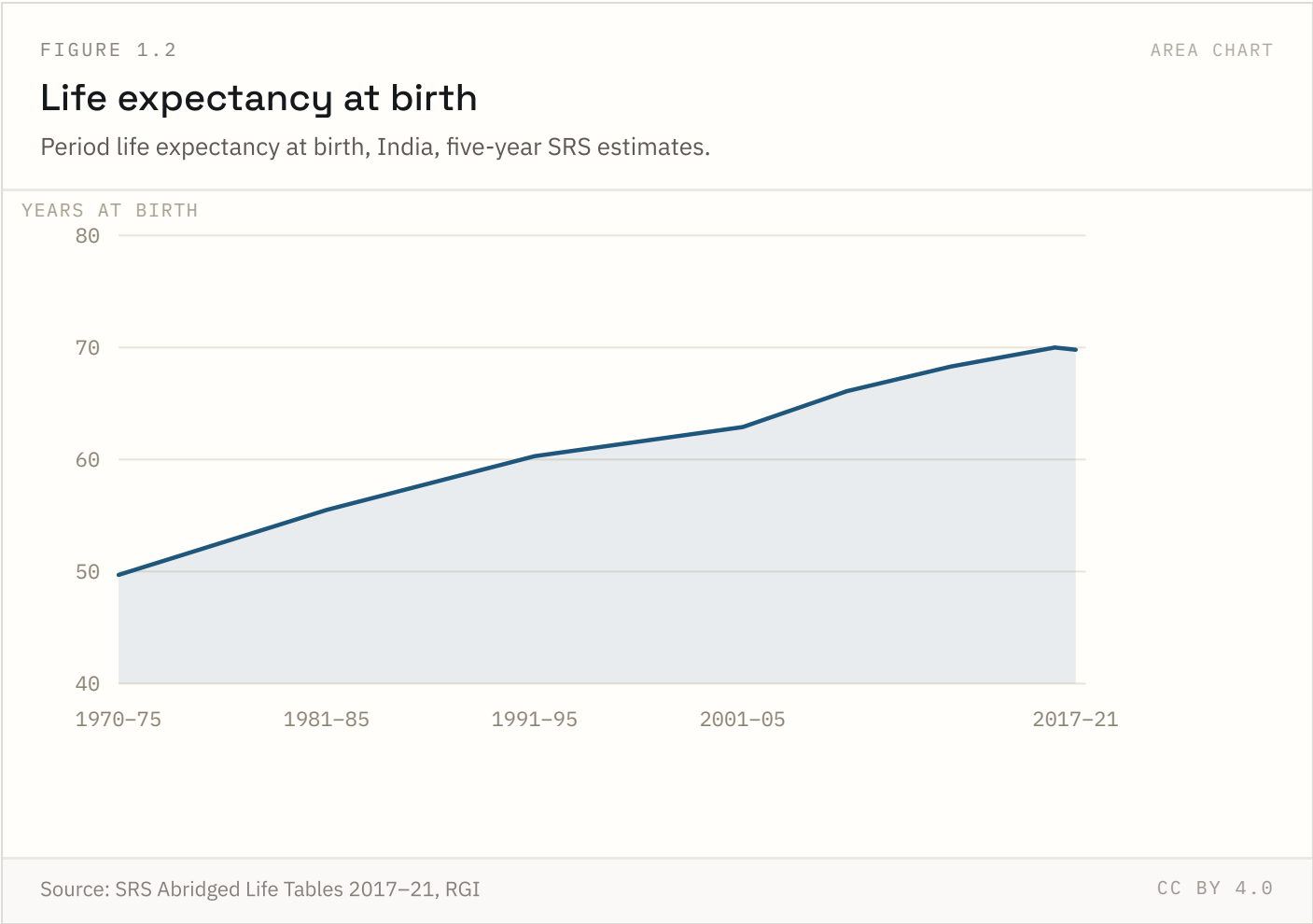
# Population & demography

4 of 10 indicators in this chapter carry state-level detail; the rest are published at national level only.

FIGURE 1.1 NATIONAL + STATE



The total fertility rate is the number of children a woman would bear if she experienced current age-specific fertility rates throughout her reproductive life. We splice decennial Census-based estimates to 1991 with the annual Sample Registration System thereafter; NFHS rounds are shown where they differ by more than 0.1.



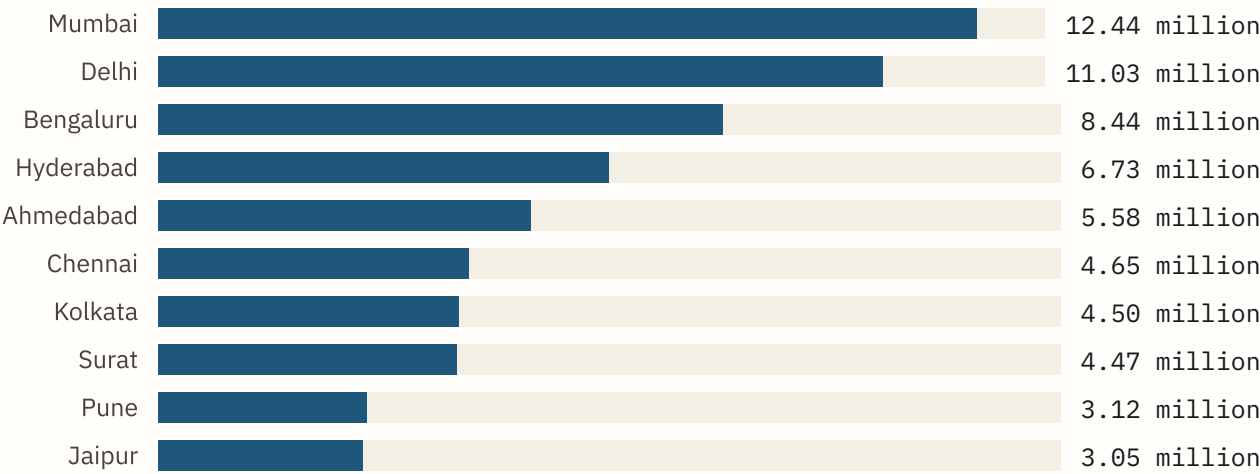
Period life expectancy at birth assumes a newborn experiences current mortality rates at every age for life. It is not a forecast of any individual cohort. The 2021 fall is real and reflects pandemic excess mortality; estimates of its size still vary across sources by several years.

FIGURE 1.3

RANKED BARS

## Ten largest cities by population

Population within municipal limits (city proper), excluding the wider urban agglomeration.



SCALE 0 - 13.50 MILLION

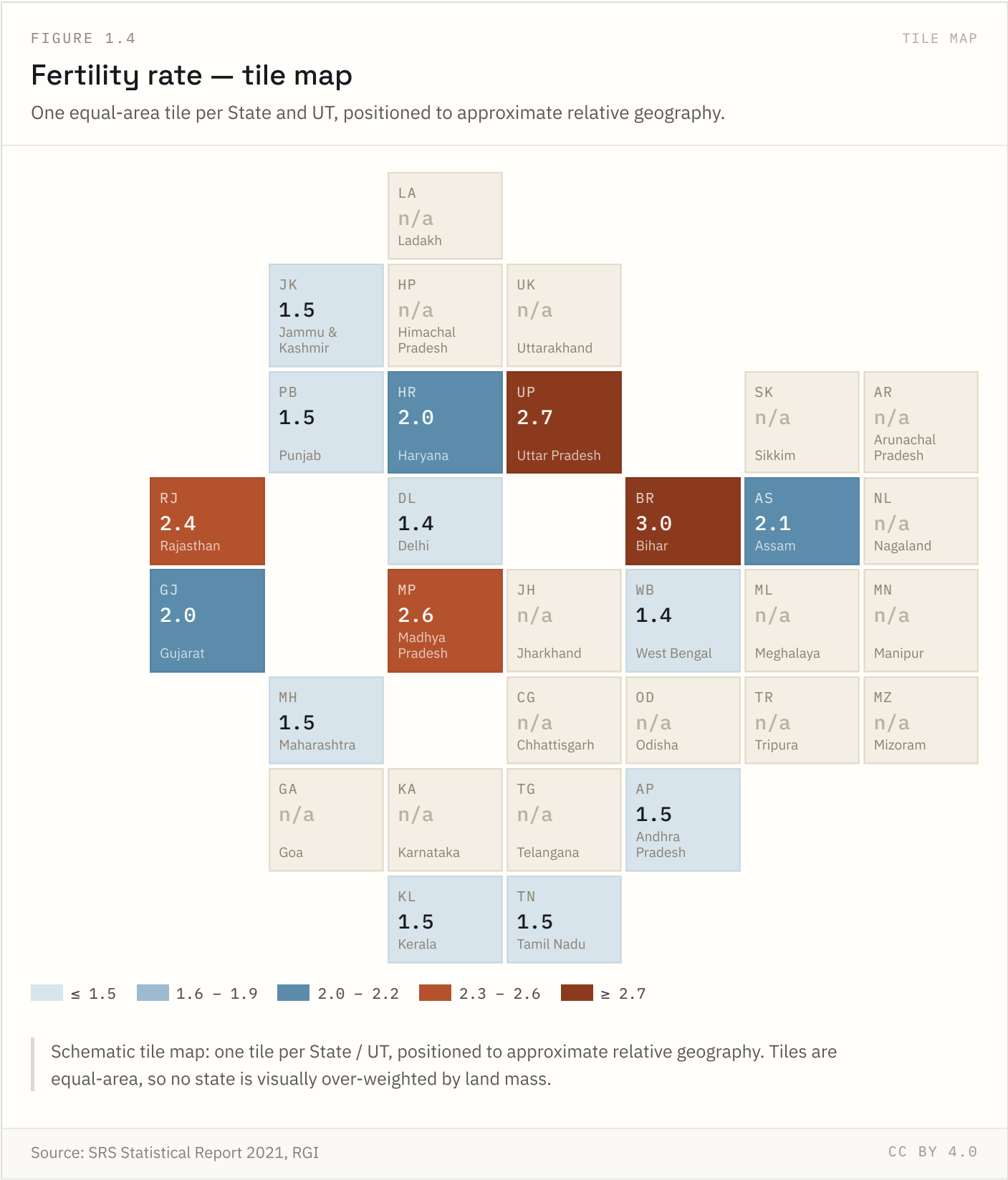
City-proper population, within municipal limits — smaller than the wider urban agglomeration. Delhi's urban agglomeration (16.3 million) is roughly 48% larger than the city figure shown here.

Source: Census of India 2011

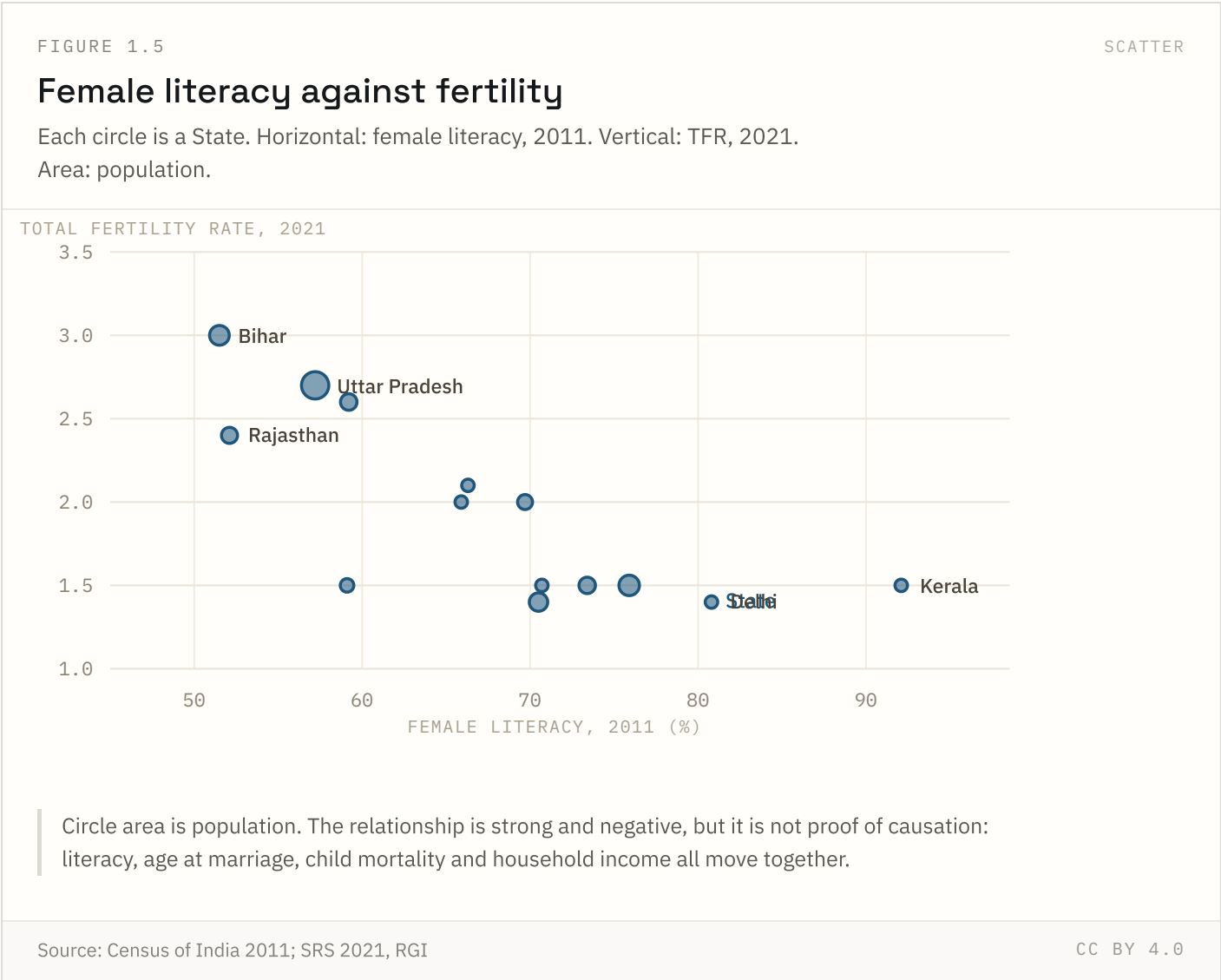
CC BY 4.0

City-proper population excludes adjoining outgrowths and satellite towns, so it understates real metropolitan size — Delhi's urban agglomeration (16.3 million) is roughly 48% larger than its city-proper figure. We use city-proper because it is the more consistently defined boundary across cities.

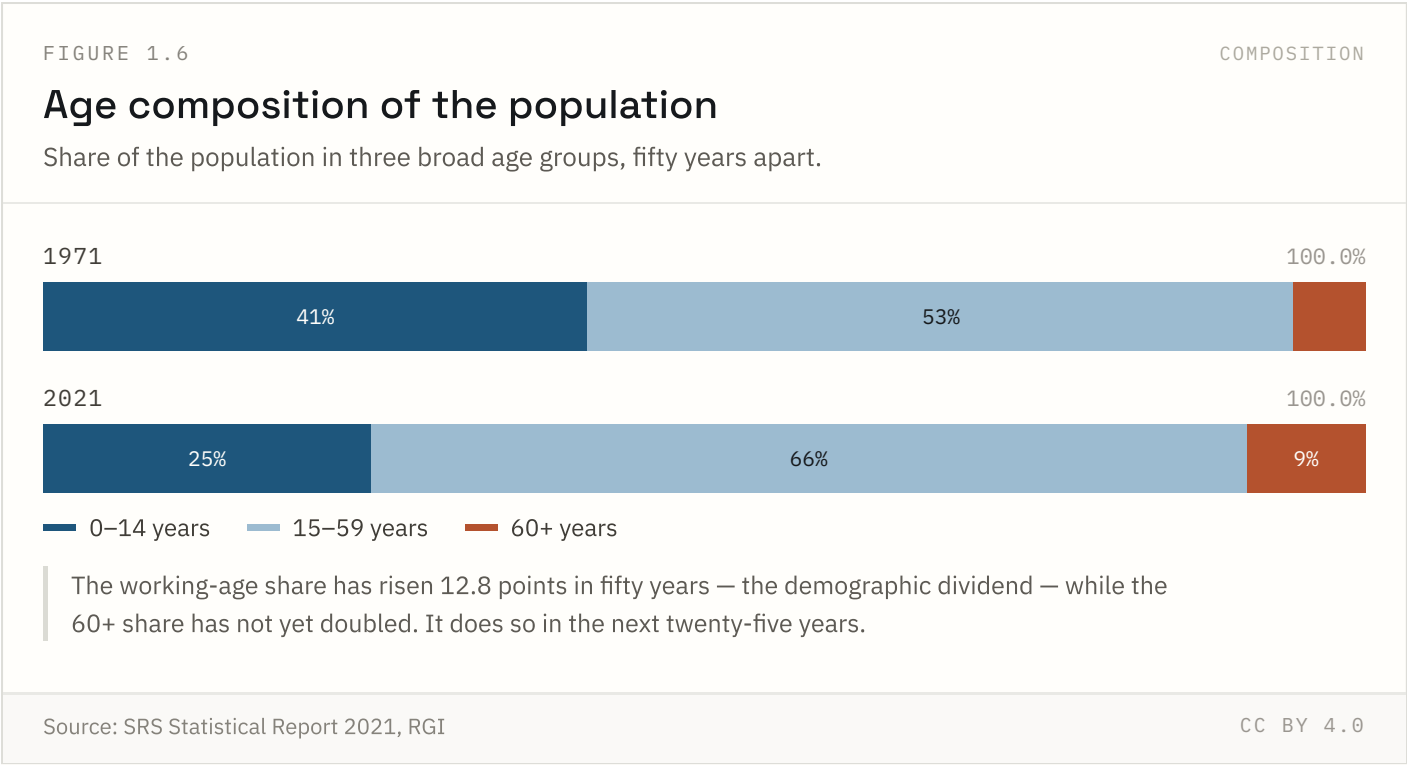
Source: Census of India 2011 Coverage: 2011 Updated: 16 Aug 2026



A tile map removes the visual bias of land area: Rajasthan does not shout louder than Kerala. Tiles marked n/a are States and UTs for which the SRS 2021 report does not publish a separate TFR estimate.



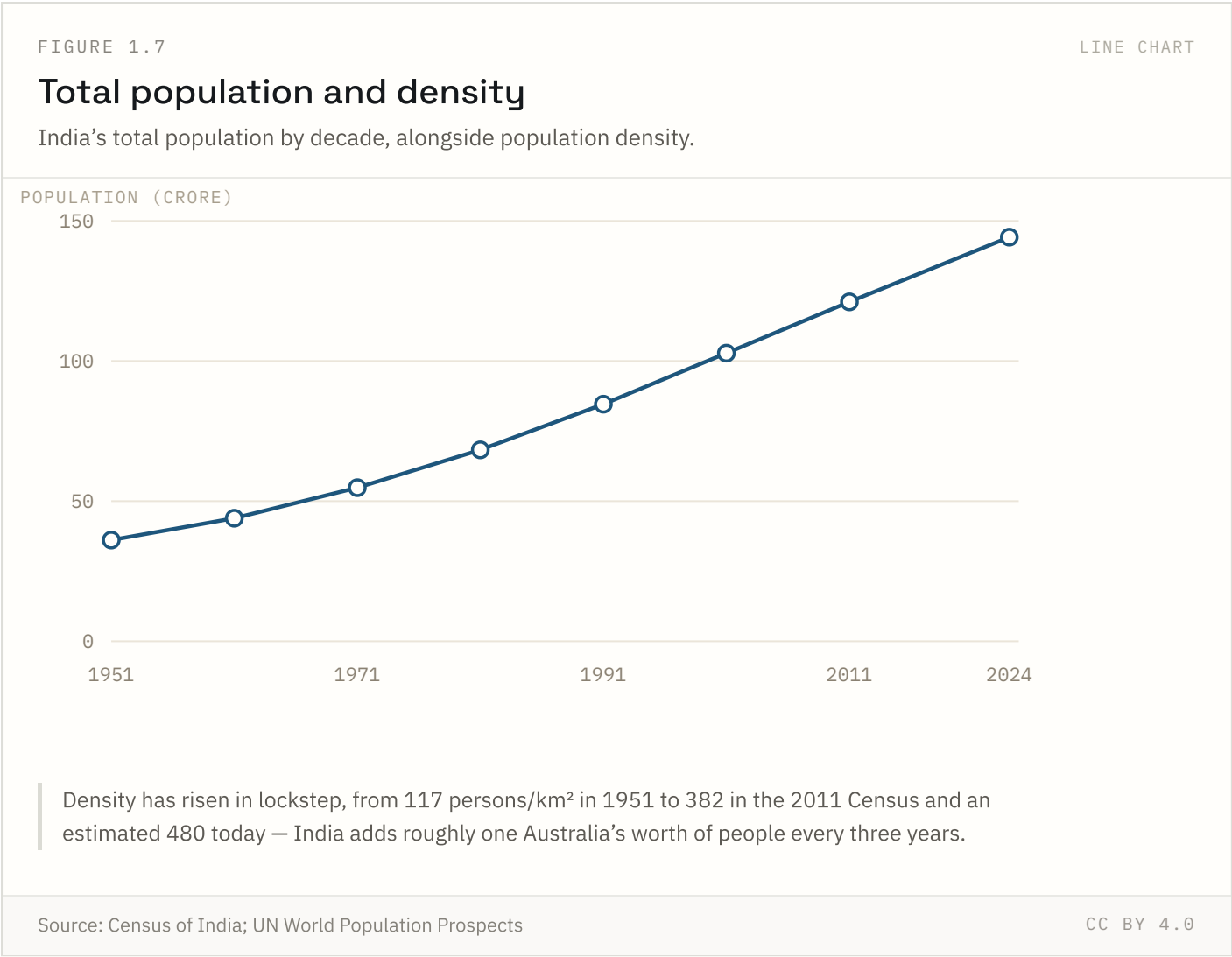
The negative relationship is strong across States, but it is not evidence of causation on its own — literacy, age at marriage, child mortality and household income all move together. Read it as a description of the fertility map, not a mechanism.



The 0–14 share fell from 41.2% to 24.8% between 1971 and 2021 while the working-age share rose to 66.2%. That gap is the demographic dividend; it closes as the 60+ share, now 9.0%, roughly doubles by 2050.



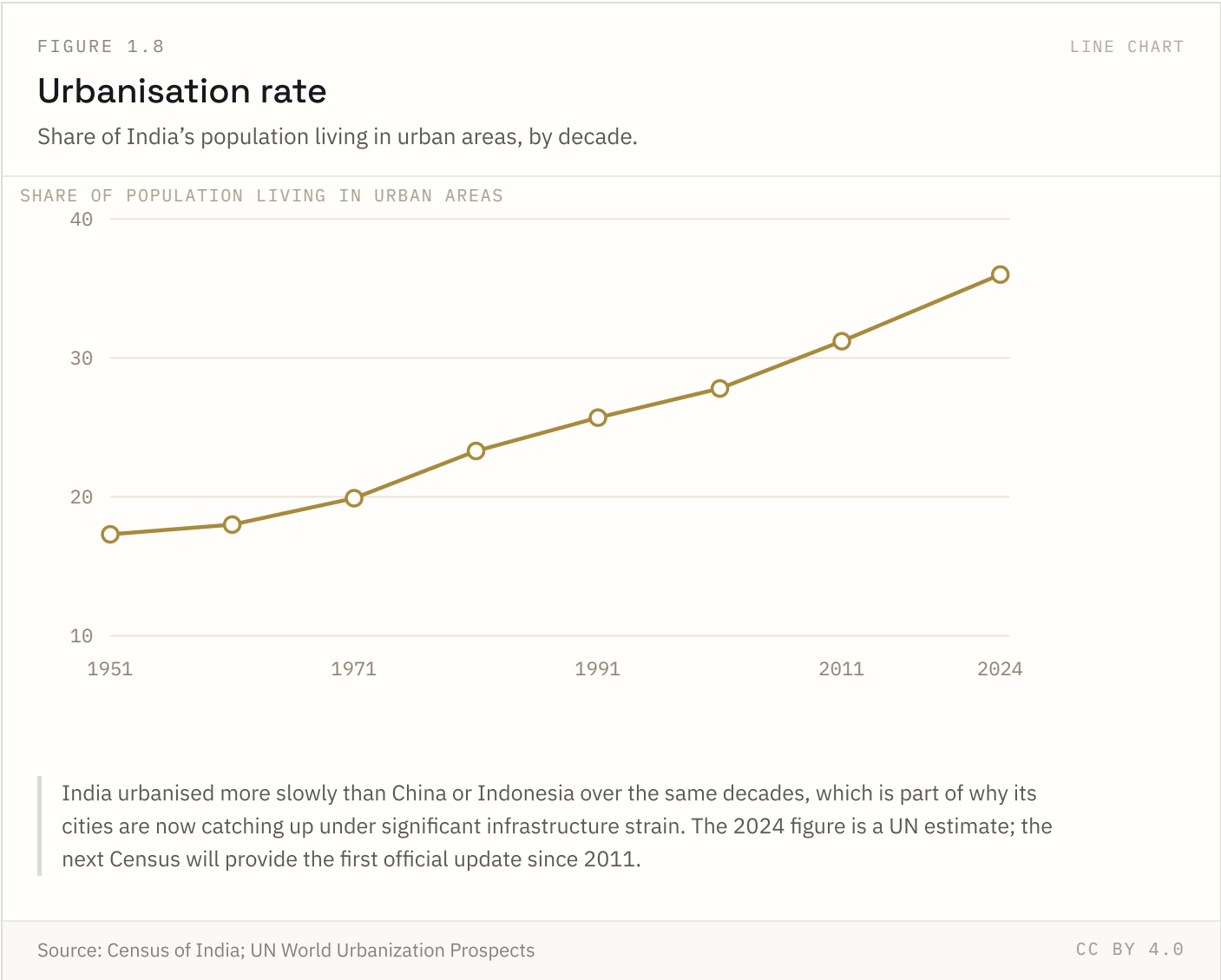
FIGURE 1.7 NATIONAL ONLY



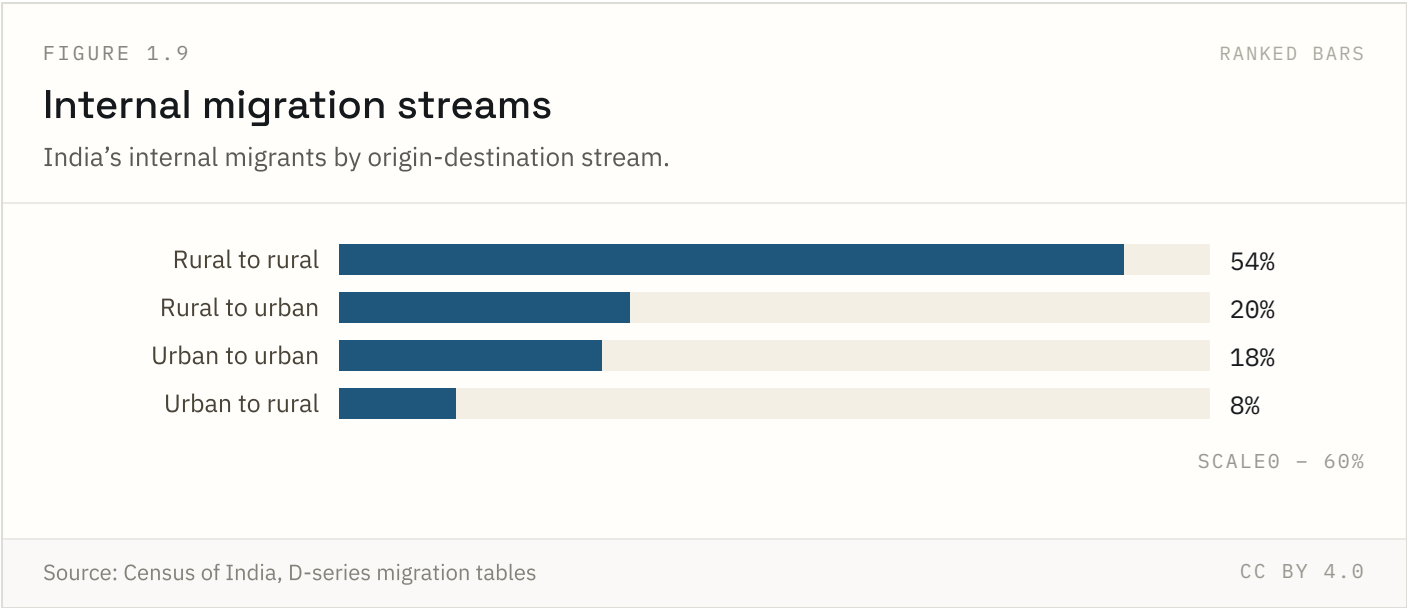
India crossed 1 billion people around 2000 and an estimated 1.44 billion by 2024, having overtaken China as the world's most populous country in 2023. Density figures assume a fixed land area, so they track population growth almost exactly.

Source: Census of India; UN World Population Prospects   Coverage: 1951–2024   Updated: 18 Aug 2026

FIGURE 1.8 NATIONAL ONLY

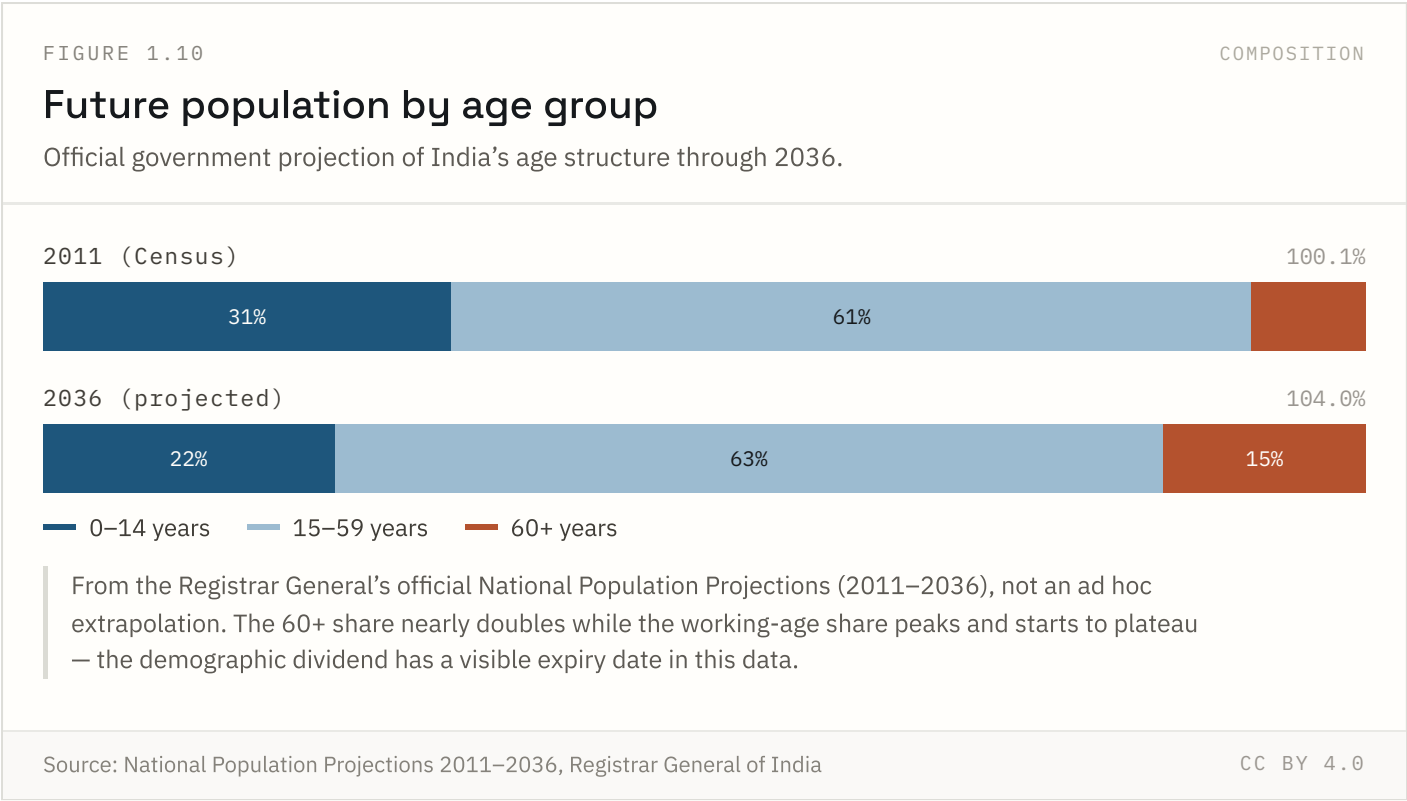


India’s urbanisation has been notably gradual compared to East Asian peers — a mixed blessing that avoided some of the informal-settlement crises seen elsewhere, but also means the next wave of urban growth, now underway, is arriving on infrastructure built for a smaller urban population.



Rural-to-rural migration, mostly marriage-related moves by women, is the largest stream by volume — rural-to-urban labour migration, the stream most associated with India’s growth story, is actually the second largest, not the biggest.

Source: Census of India, D-series migration tables   Coverage: 2011   Updated: 18 Aug 2026



This is RGI's own published projection, not an extrapolation built for this site. It shows the demographic dividend has a visible expiry date: the working-age share is projected to plateau around the mid-2030s as the 60+ share nearly doubles from its 2011 level.

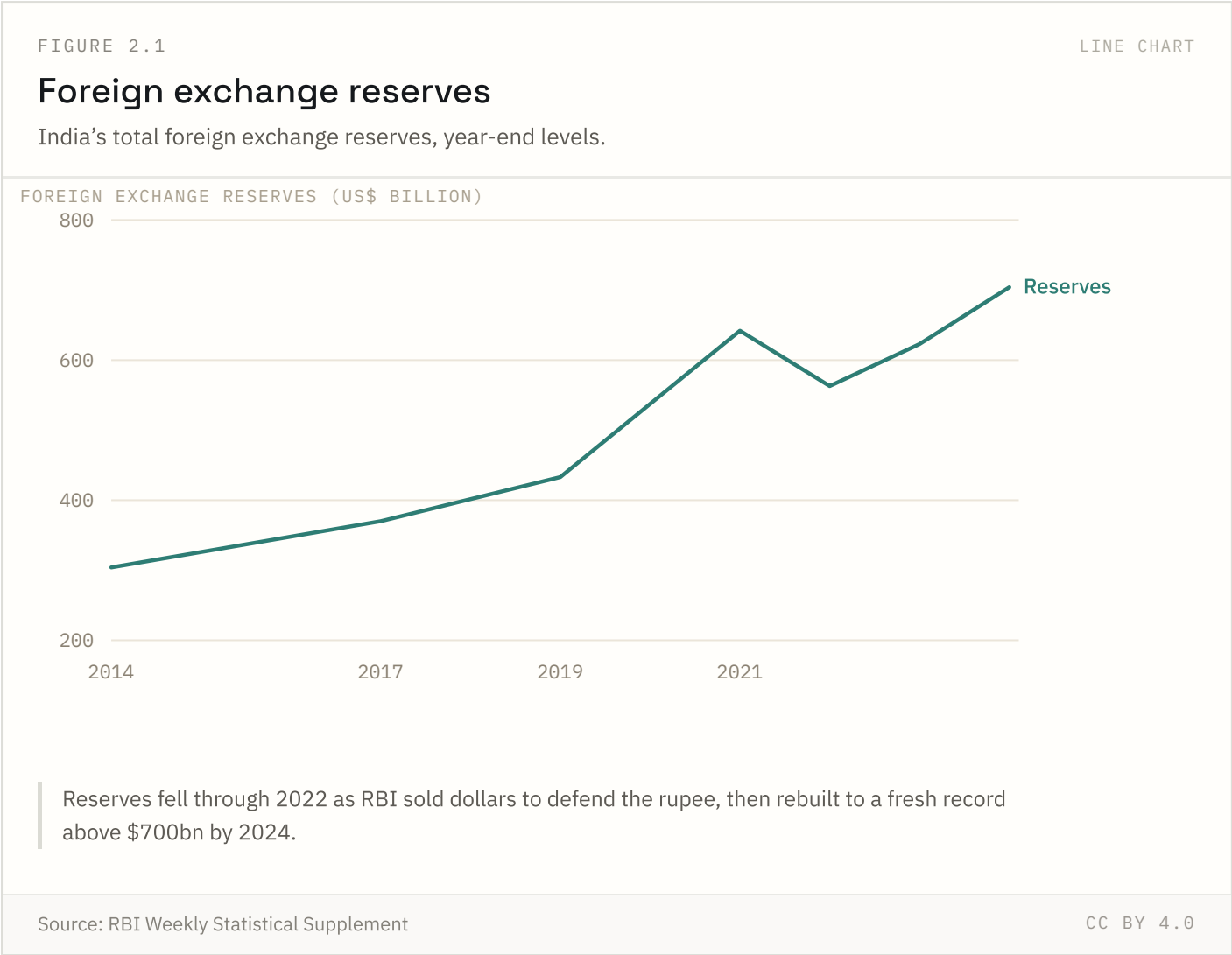
Source: National Population Projections 2011–2036, Registrar General of India

Coverage: 2011 to 2036 (projected)    Updated: 18 Aug 2026

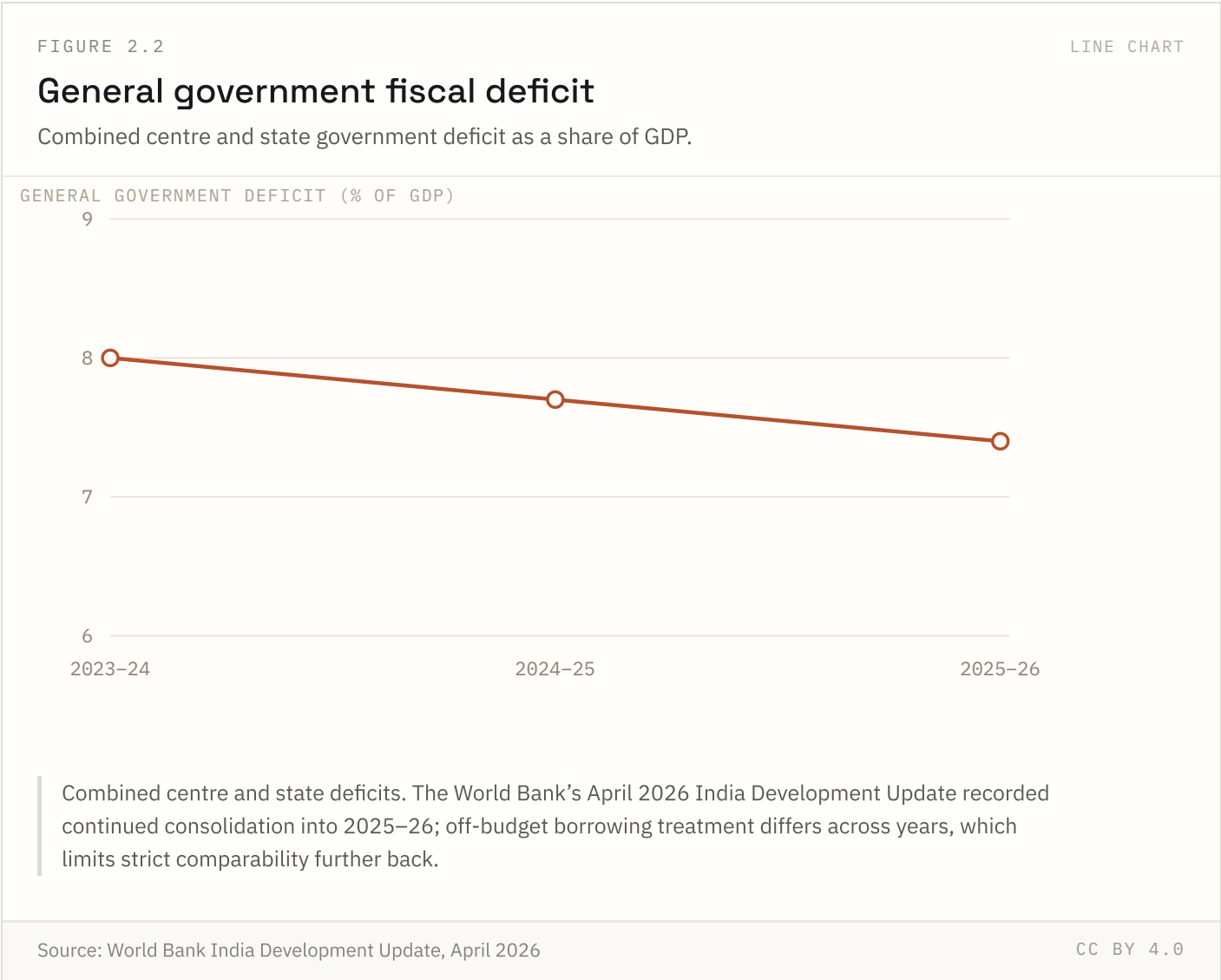
# Economy & finance

1 of 13 indicators in this chapter carry state-level detail; the rest are published at national level only.

FIGURE 2.1 NATIONAL ONLY

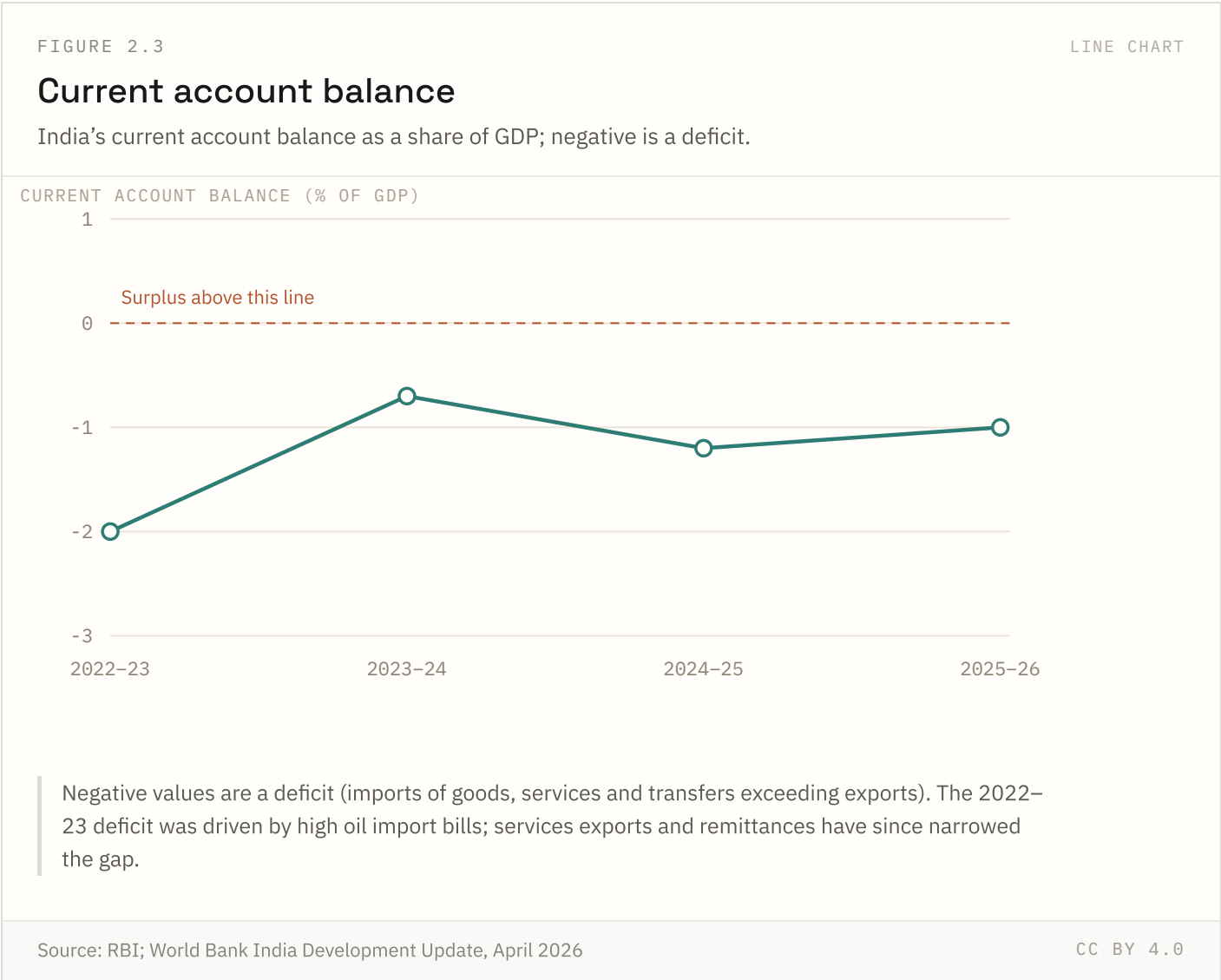


Reserves fell through 2022 as the RBI sold dollars to slow the rupee’s depreciation, then rebuilt past \$700 billion by 2024 — among the largest reserve stockpiles of any country.



The deficit has narrowed for three straight years as the centre held to its fiscal consolidation path and states’ combined deficit stayed stable. Off-budget borrowing is treated inconsistently across years and agencies, which can shift the level without a real change in underlying fiscal stance.

FIGURE 2.3 NATIONAL ONLY

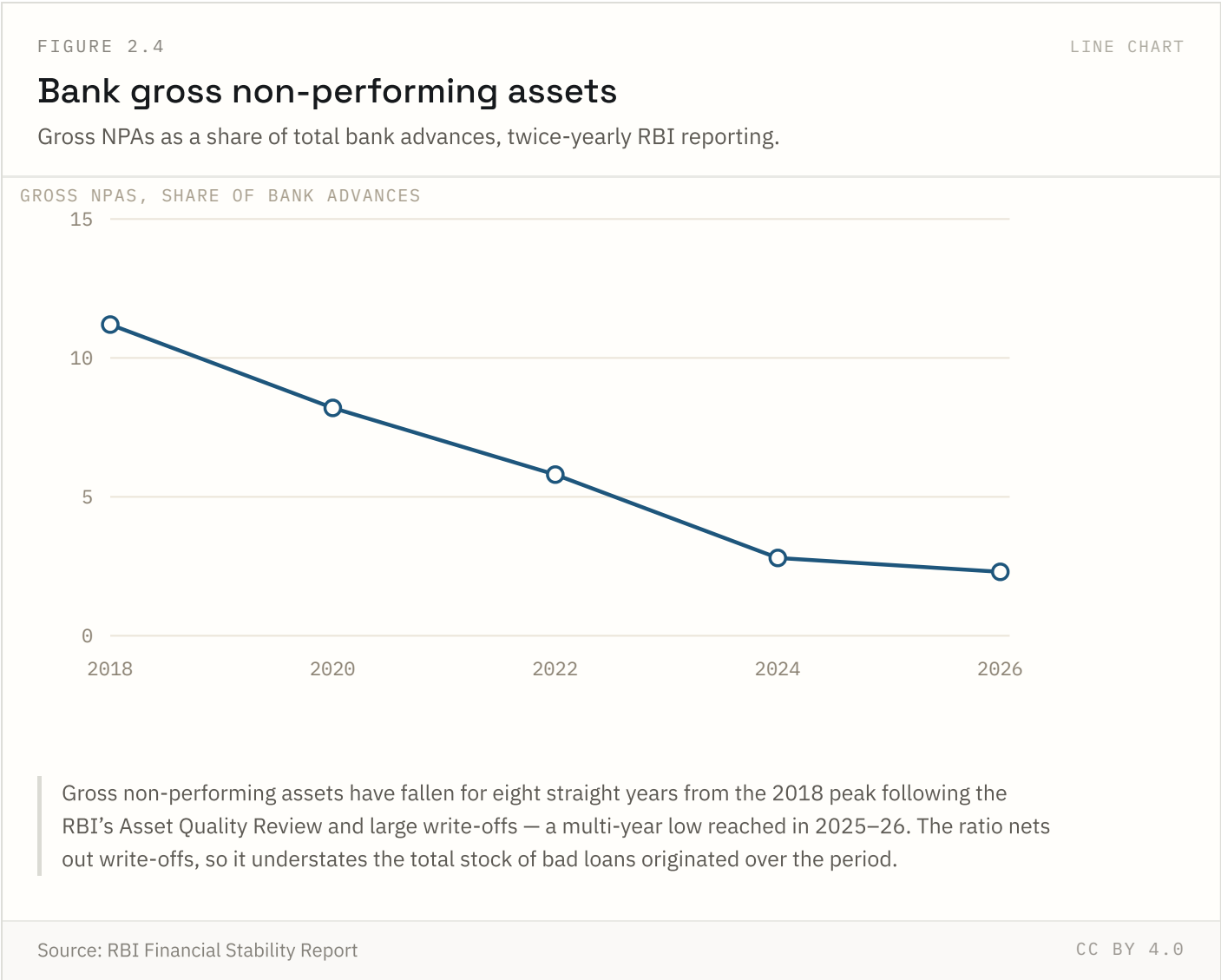


The 2022–23 deficit peaked on a high oil import bill; strong services exports and remittance inflows have since kept the deficit comfortably financeable. A current account deficit is normal for a fast-growing economy importing capital goods and energy.

Source: RBI; World Bank India Development Update, April 2026 Coverage: 2022–23 to 2025–26

Updated: 18 Aug 2026

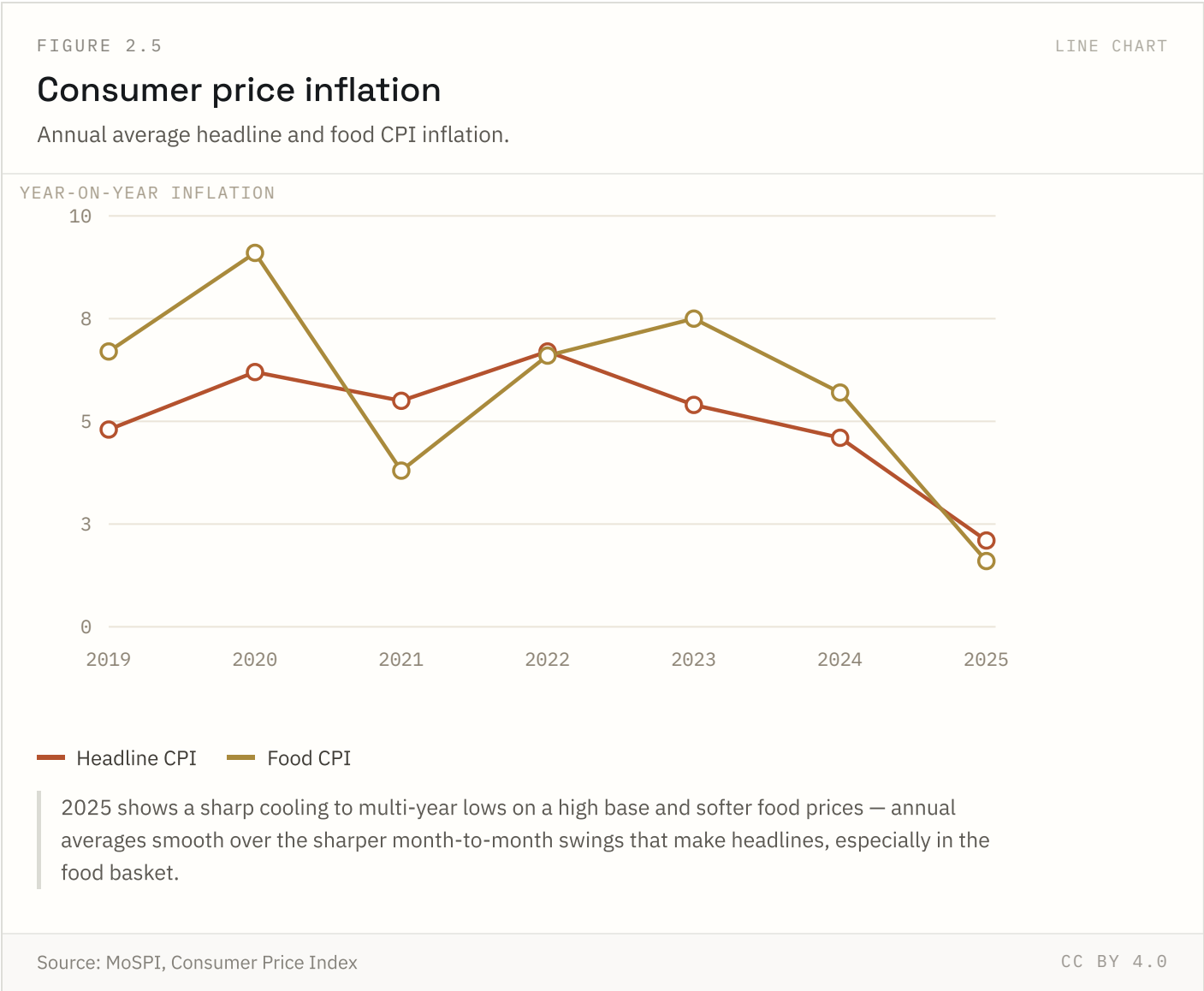
FIGURE 2.4 NATIONAL ONLY



The ratio has fallen every year since the 2018 peak that followed the RBI’s Asset Quality Review, helped by large write-offs and improved recovery under the Insolvency and Bankruptcy Code. A falling ratio reflects both fewer new bad loans and banks removing old ones from the book, so it is not a pure flow measure.

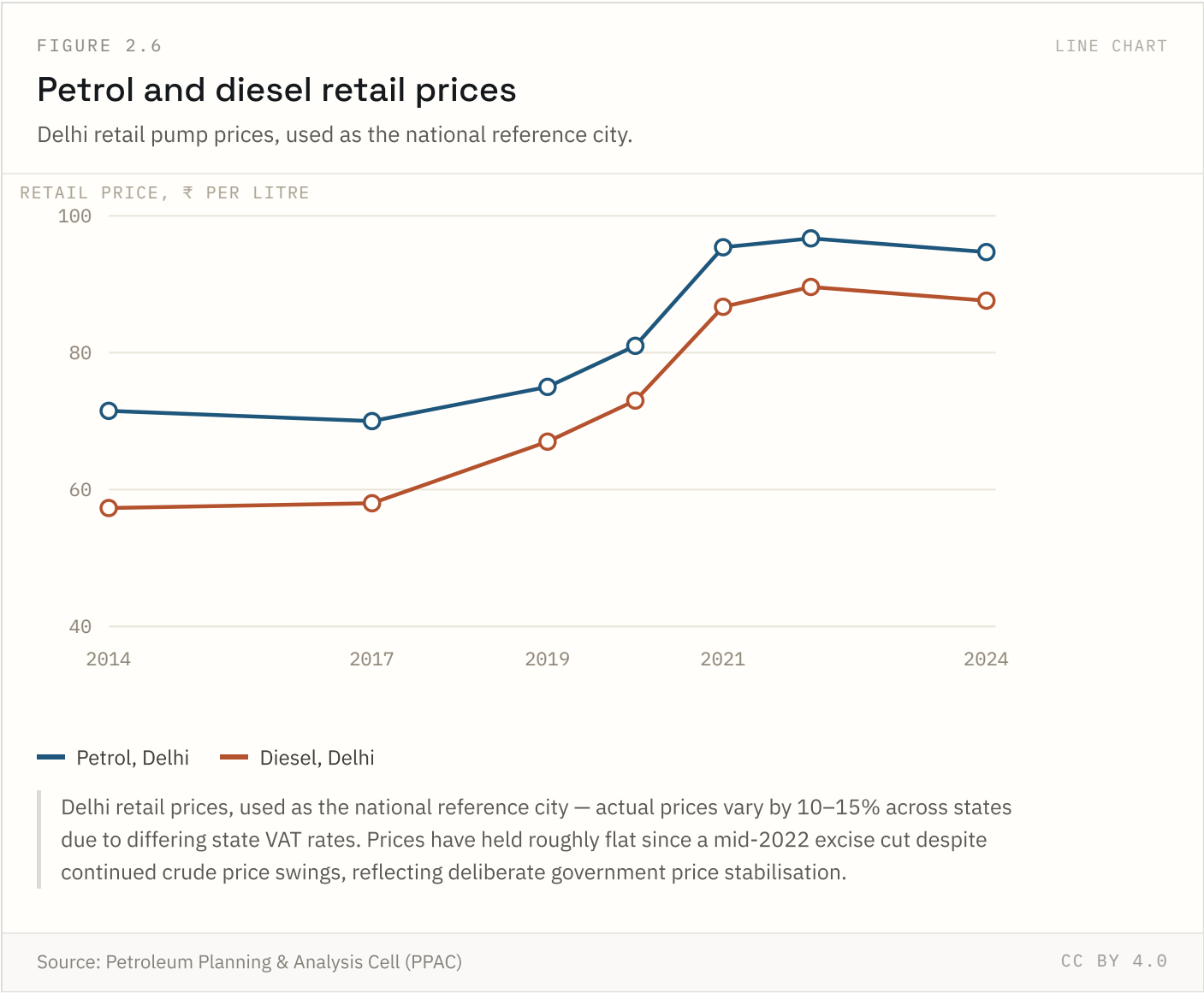


FIGURE 2.5 NATIONAL ONLY



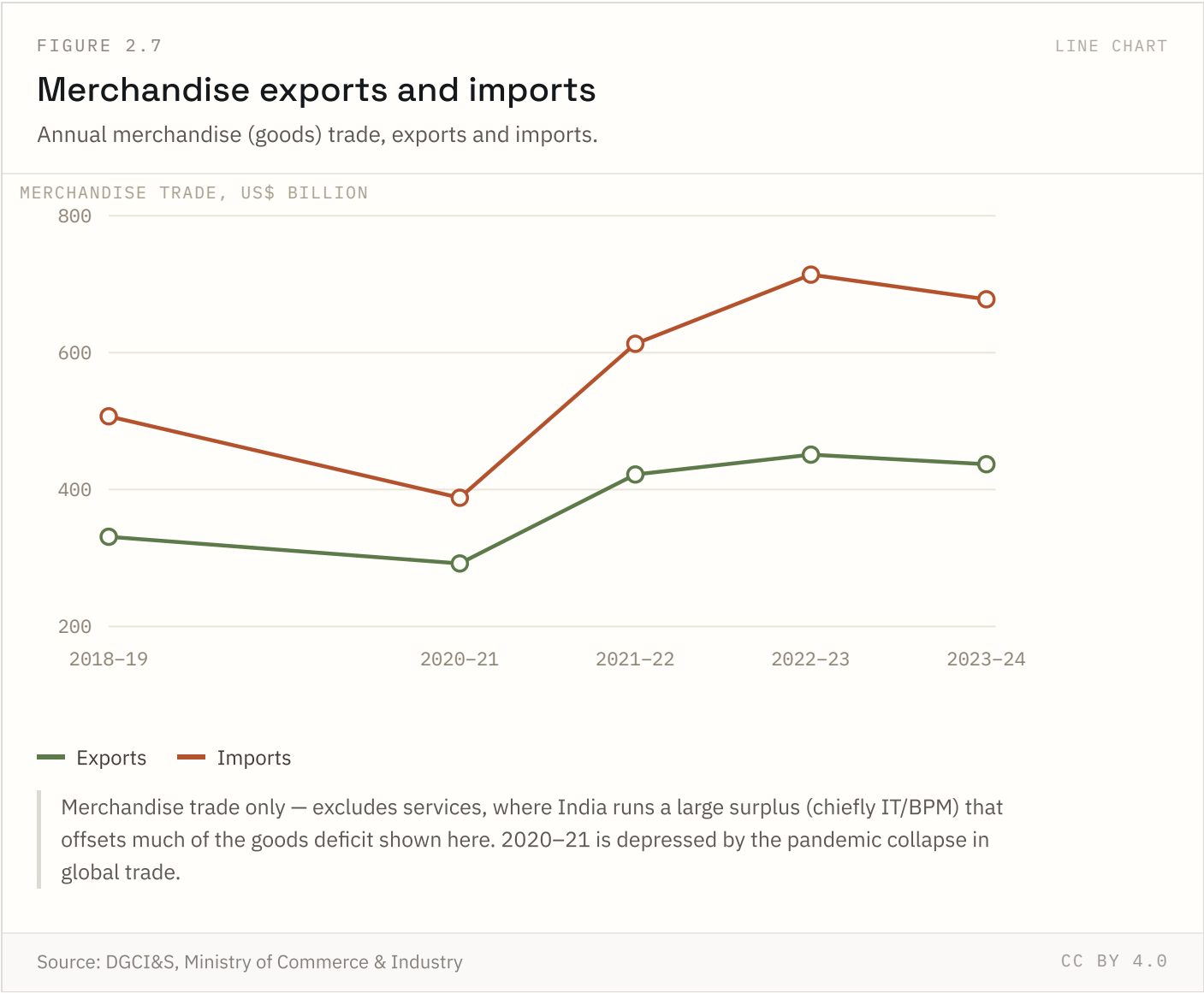
Food inflation runs more volatile than the headline rate in both directions, since it is exposed to monsoon shocks and global commodity swings that core inflation is not. Annual averages mask sharper month-to-month spikes, particularly around vegetable price shocks.

FIGURE 2.6 NATIONAL ONLY

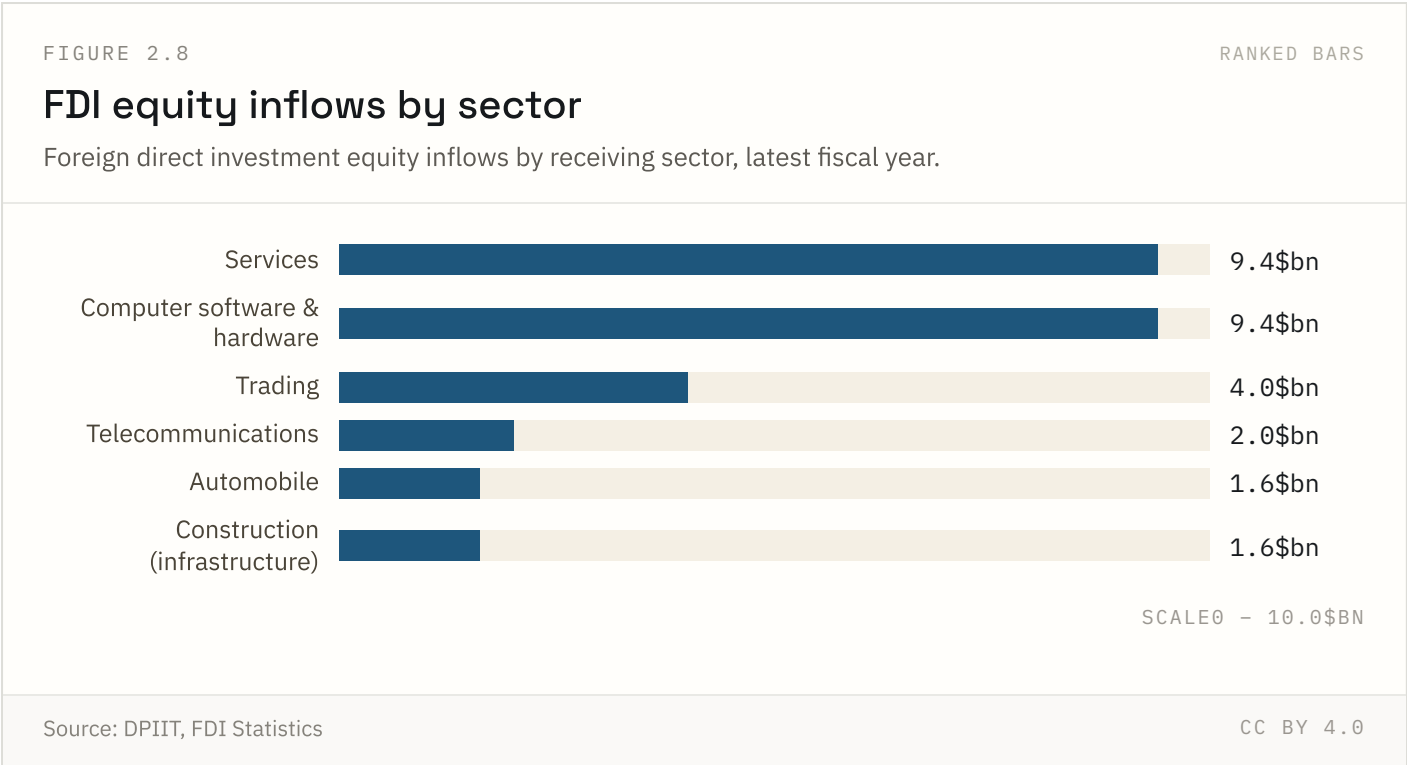


Retail prices are a mix of global crude prices, the rupee exchange rate, and state and central taxes — taxes make up roughly 40% of the pump price. Prices have barely moved since mid-2022 despite crude volatility, a deliberate stabilisation choice rather than a market outcome.

FIGURE 2.7 NATIONAL ONLY

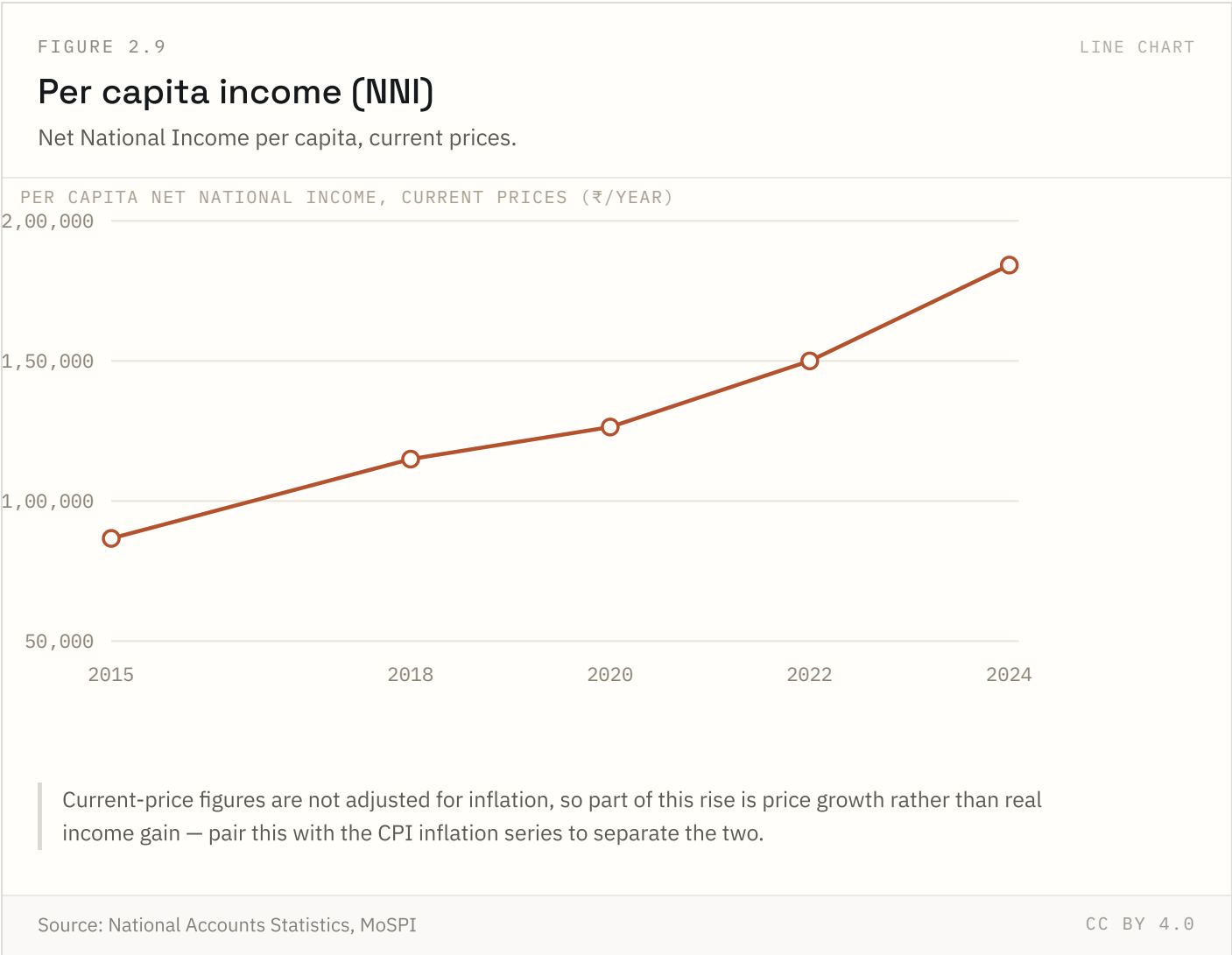


India runs a persistent merchandise trade deficit, financed in large part by a services trade surplus (led by IT/BPM exports) not shown here. The 2022-23 and 2023-24 import bills were pushed up by elevated global energy prices.

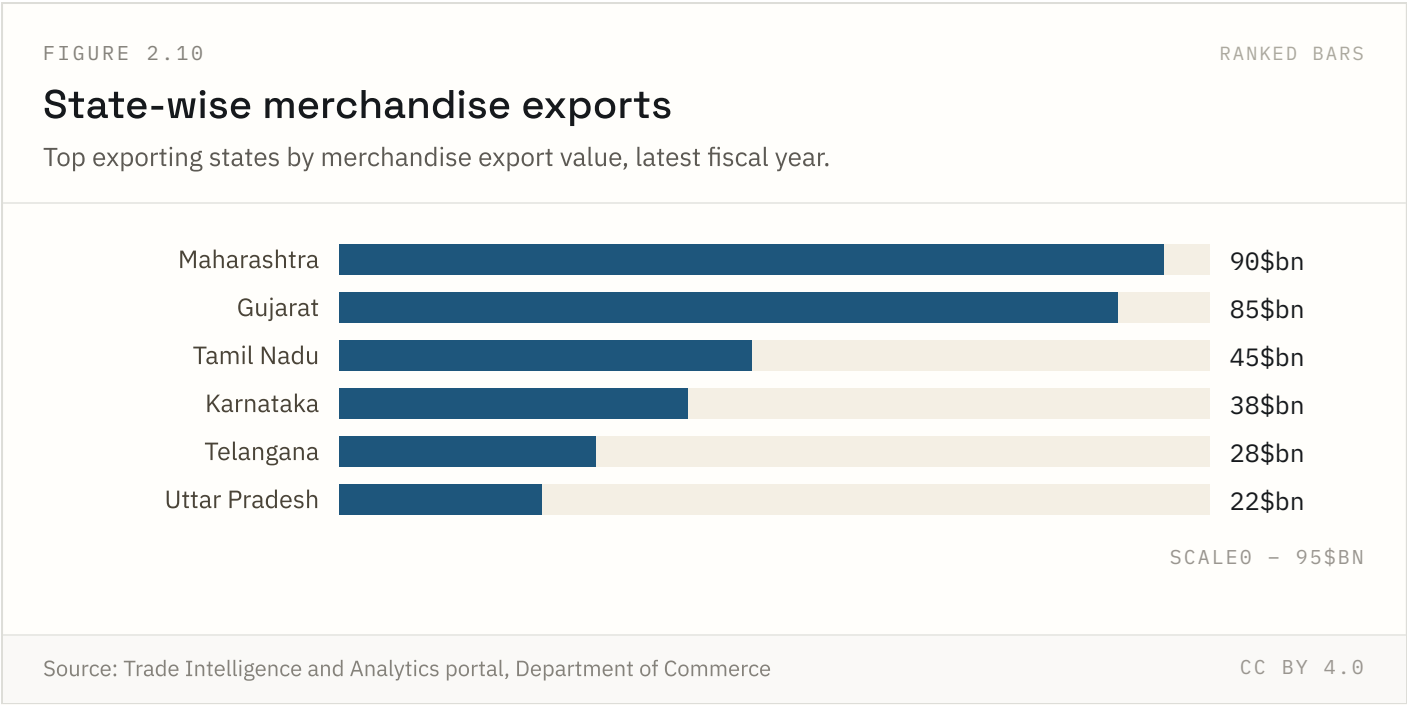


Services and computer software/hardware together draw in roughly as much FDI as the next four sectors combined, reflecting India’s pull for global capability centres and SaaS investment over manufacturing FDI. Reinvested earnings and other capital, tracked separately by DPIIT, are excluded.

FIGURE 2.9 NATIONAL ONLY

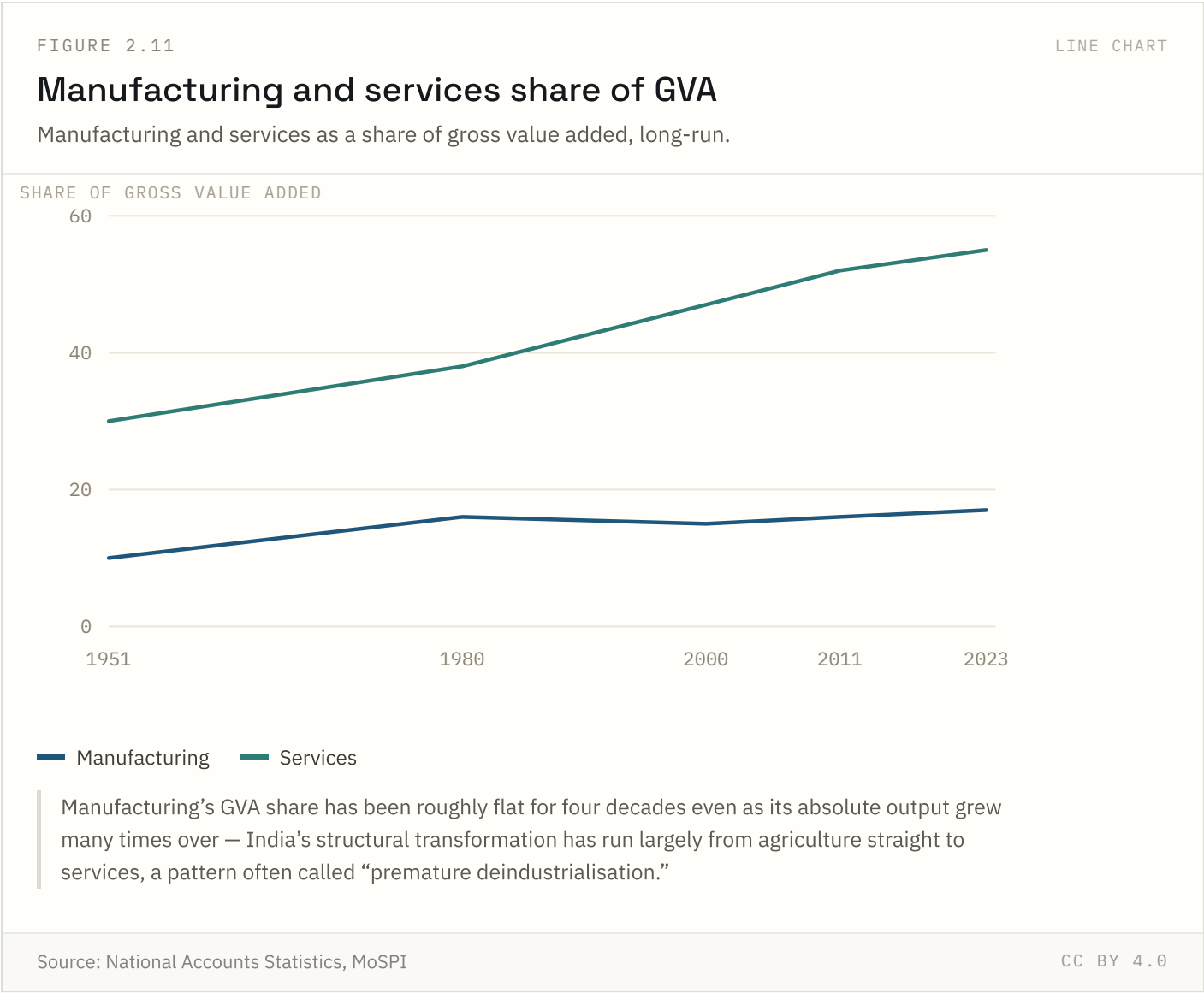


This translates aggregate GDP growth into a per-person figure, though as a national average it says nothing about distribution — pair it with the MPCE and Gini series in Poverty & incomes to see how income and consumption are actually shared.

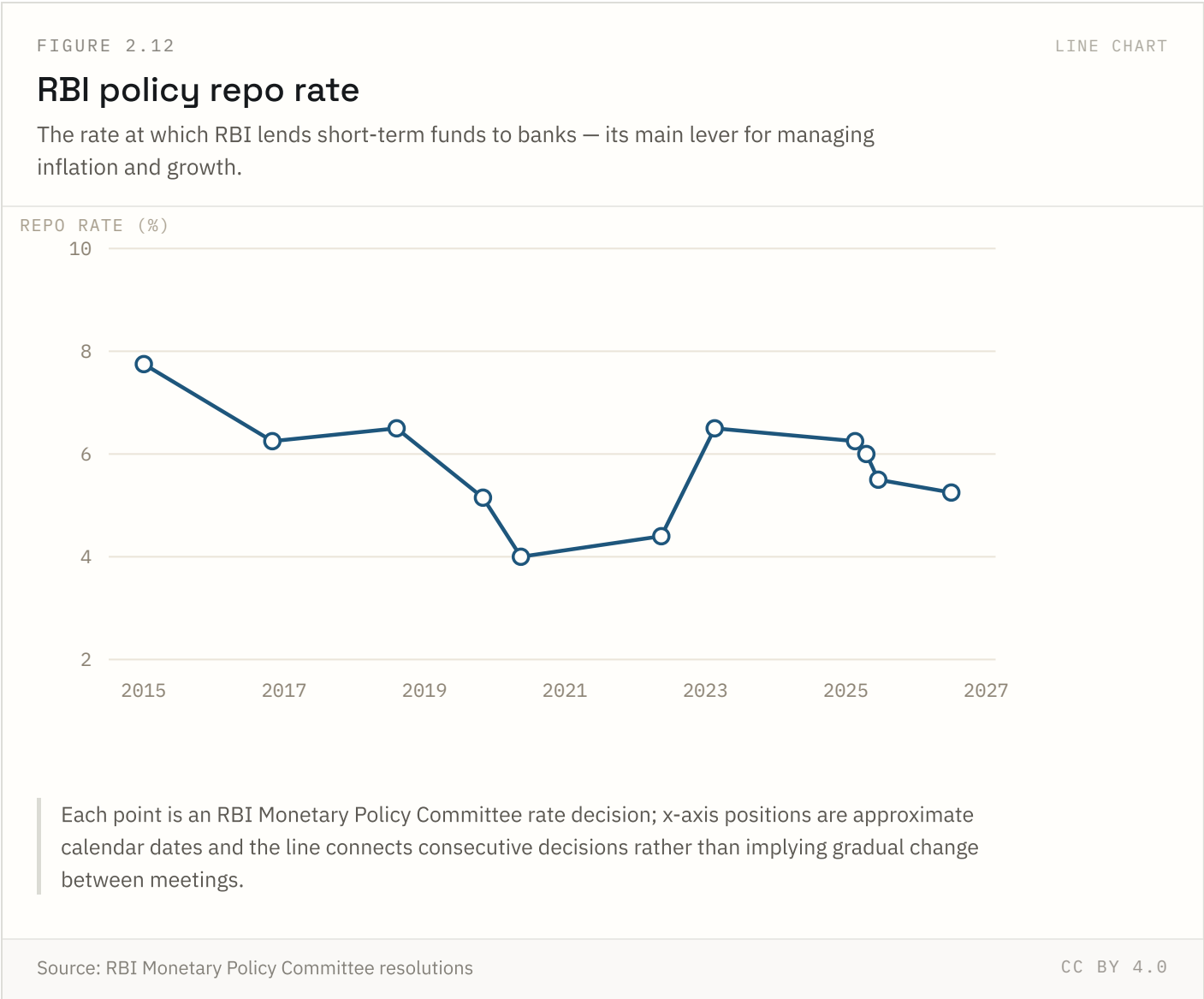


Maharashtra and Gujarat together export more than the next four states combined, reflecting concentrated port access and established manufacturing and chemicals clusters — export capacity remains far more geographically concentrated than population or GDP.

Highest: **Maharashtra** at 90\$bn. Lowest: **Uttar Pradesh** at 22\$bn.

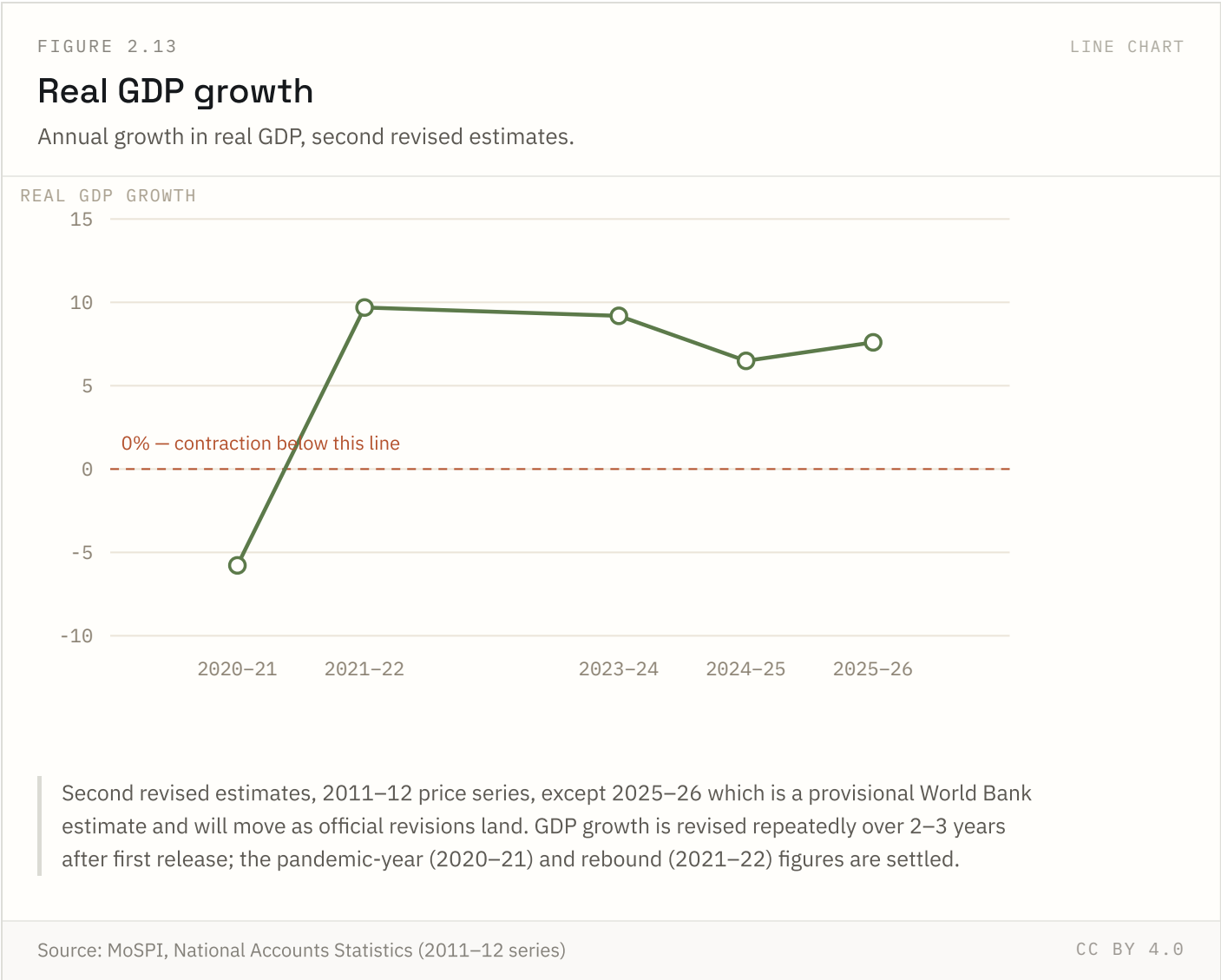


Pairs with the Agriculture & food topic’s GVA-share chart to complete the picture: India’s economy moved from agriculture-dominated to services-dominated with manufacturing’s share barely moving — a structural pattern quite different from China’s manufacturing-led transition.



Each point is an actual policy decision, not a smoothed average. The rate fell to a historic low of 4.00% during the pandemic, then rose 250 basis points across 2022–23 to fight inflation, and has eased since early 2025 as inflation moderated.





GDP estimates are revised repeatedly over two to three years after first release. The 2020–21 contraction and 2021–22 rebound are now settled figures; 2023–24 and 2024–25 may still move in future revisions.

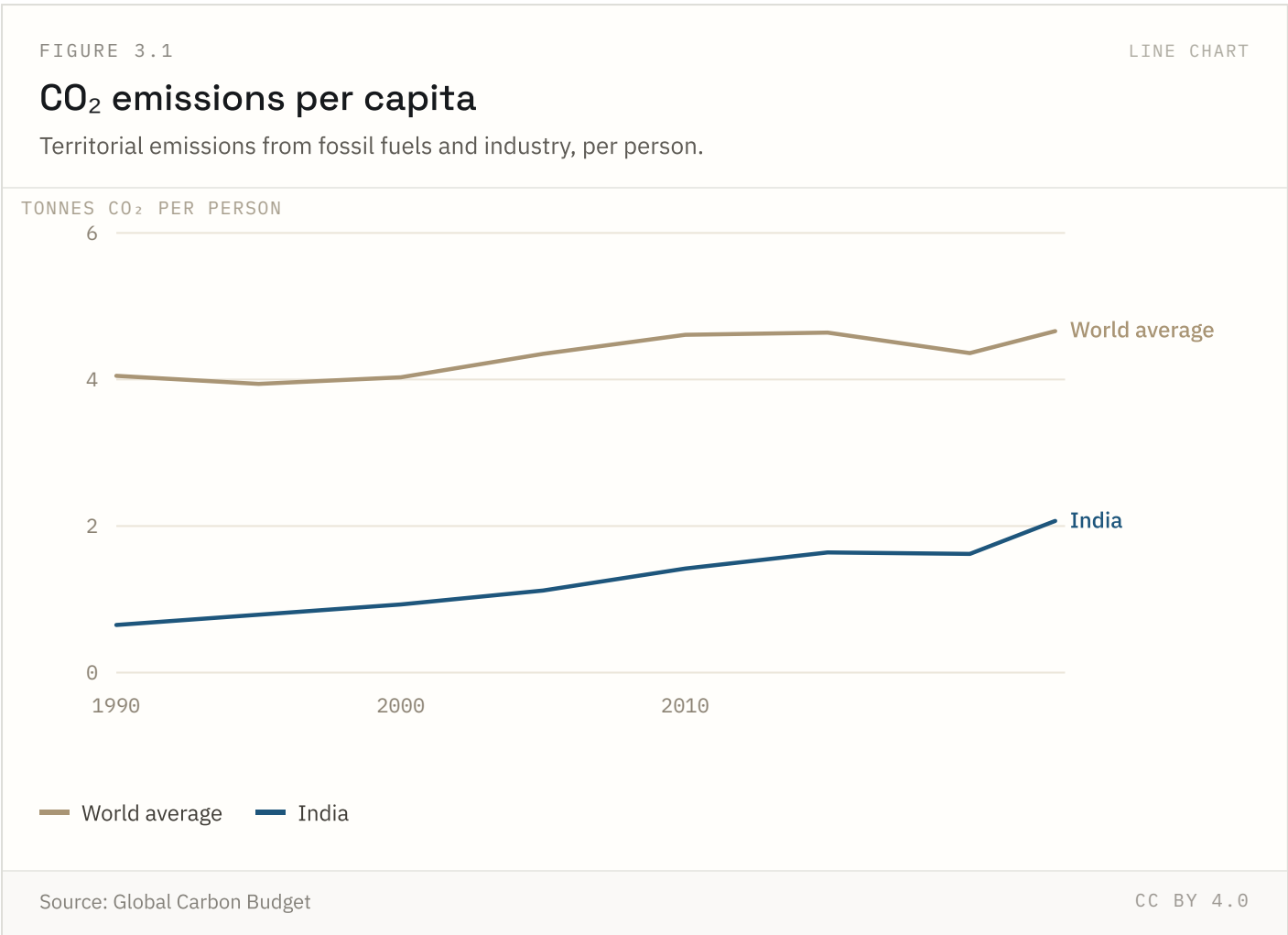
Source: MoSPI, National Accounts Statistics (2011–12 series) Coverage: 2020–21 to 2024–25

Updated: 16 Aug 2026

# Climate & air quality

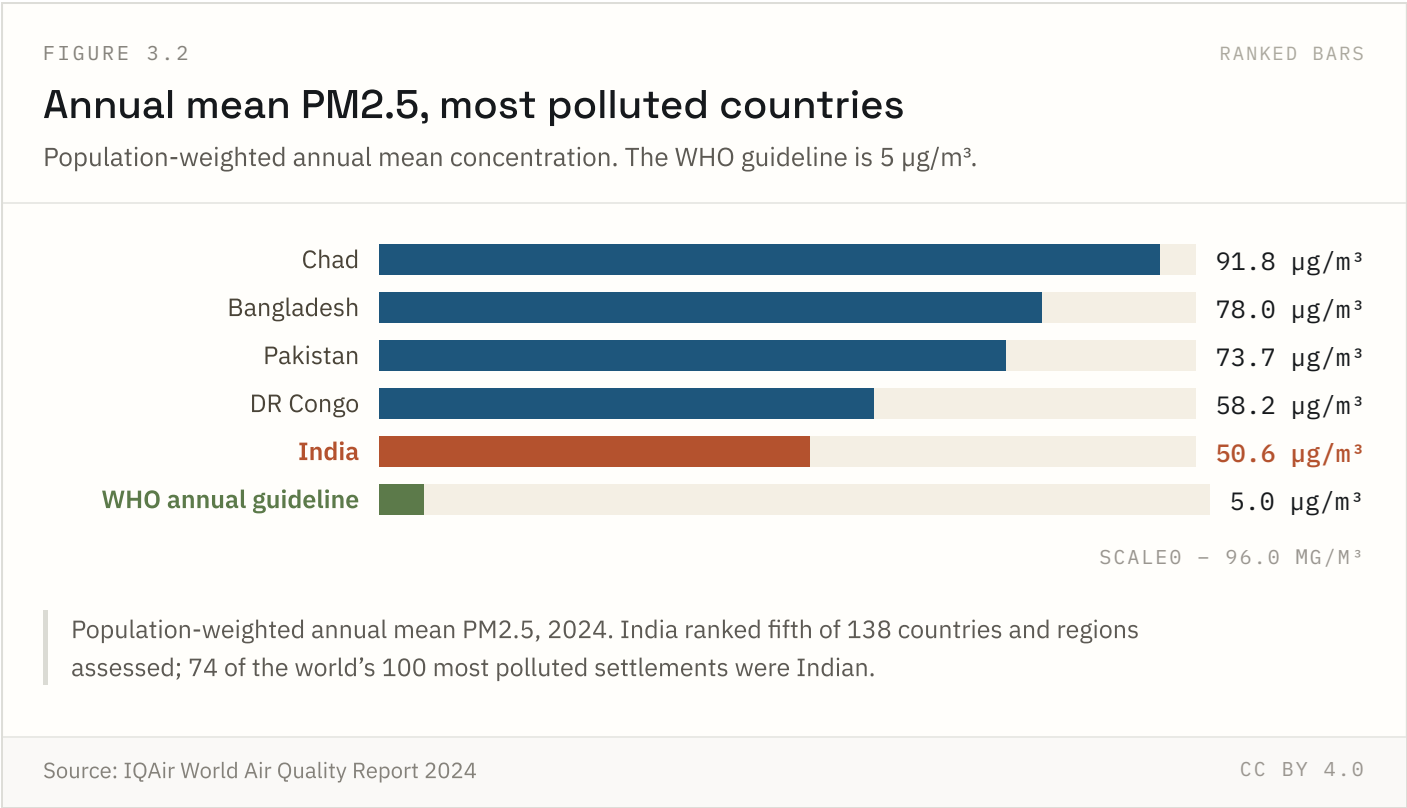
1 of 10 indicators in this chapter carry state-level detail; the rest are published at national level only.

FIGURE 3.1 NATIONAL ONLY



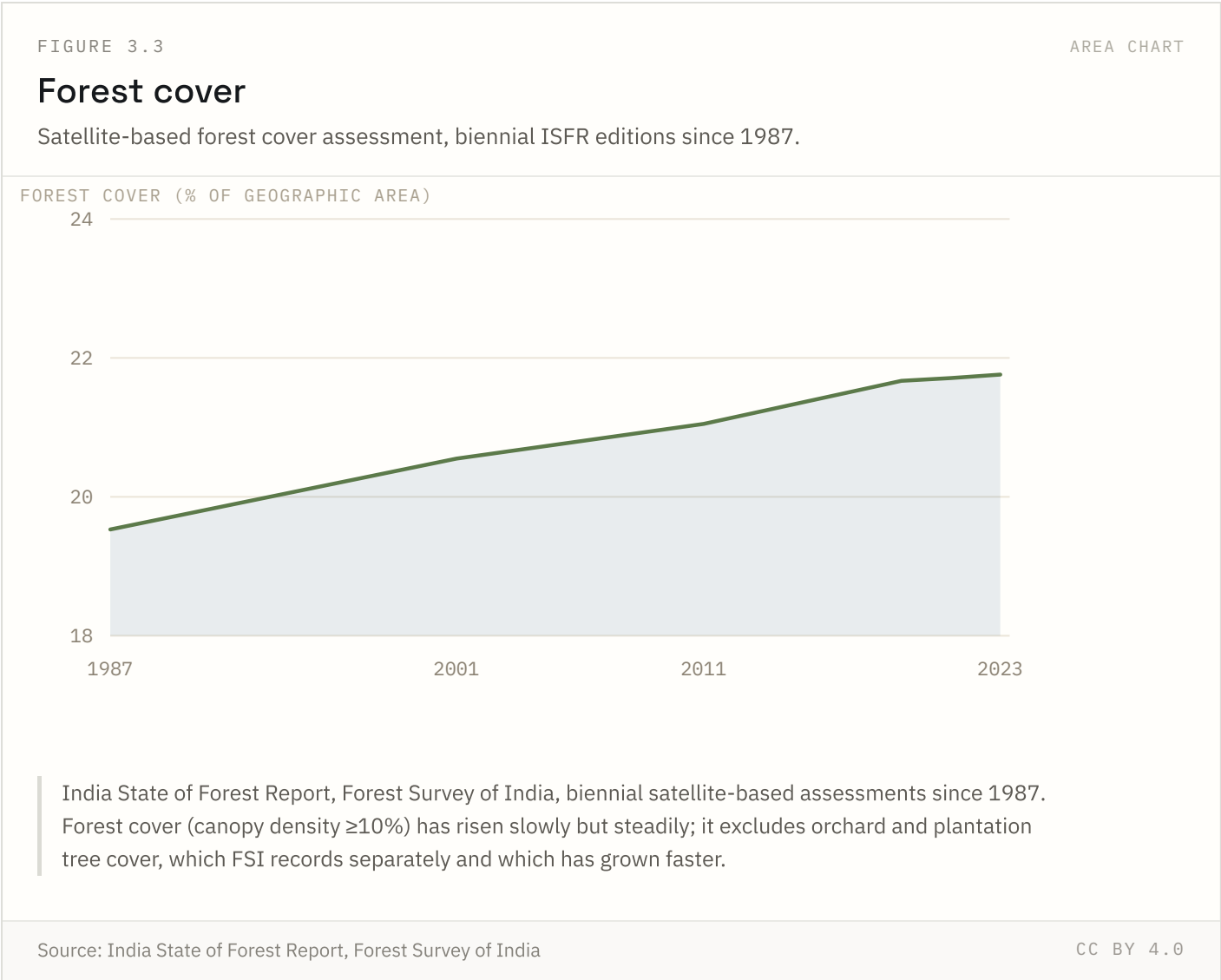
India is the third-largest emitter in absolute terms and among the lowest per person of any major economy — roughly 45 per cent of the world average. Cumulative emissions since 1850 are about 3 per cent of the global total.

FIGURE 3.2 NATIONAL ONLY



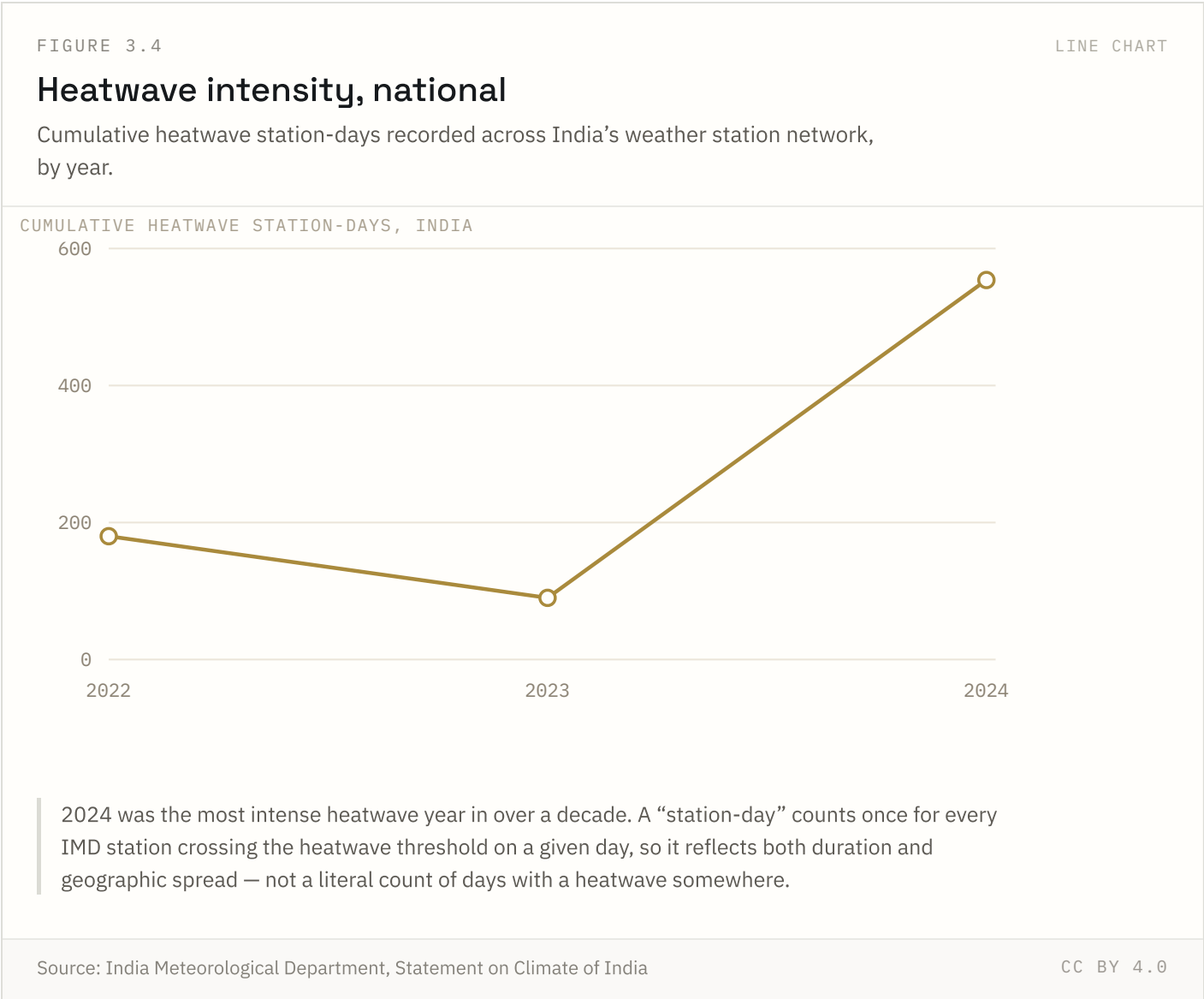
Every city shown exceeds the WHO annual guideline of 5 µg/m³, and all but one exceed India's own standard of 40. Monitor coverage is uneven: much of central and eastern India has fewer than one continuous monitor per million people.

Source: IQAir World Air Quality Report 2024 Coverage: 2024 Updated: 14 Apr 2026



Forest cover has risen slowly but steadily, from 19.53% of geographic area in 1987 to 21.76% in 2023 — still short of the National Forest Policy target of 33%. The figure excludes orchard and plantation tree cover, tracked separately, which has grown faster than natural forest.

FIGURE 3.4 NATIONAL ONLY

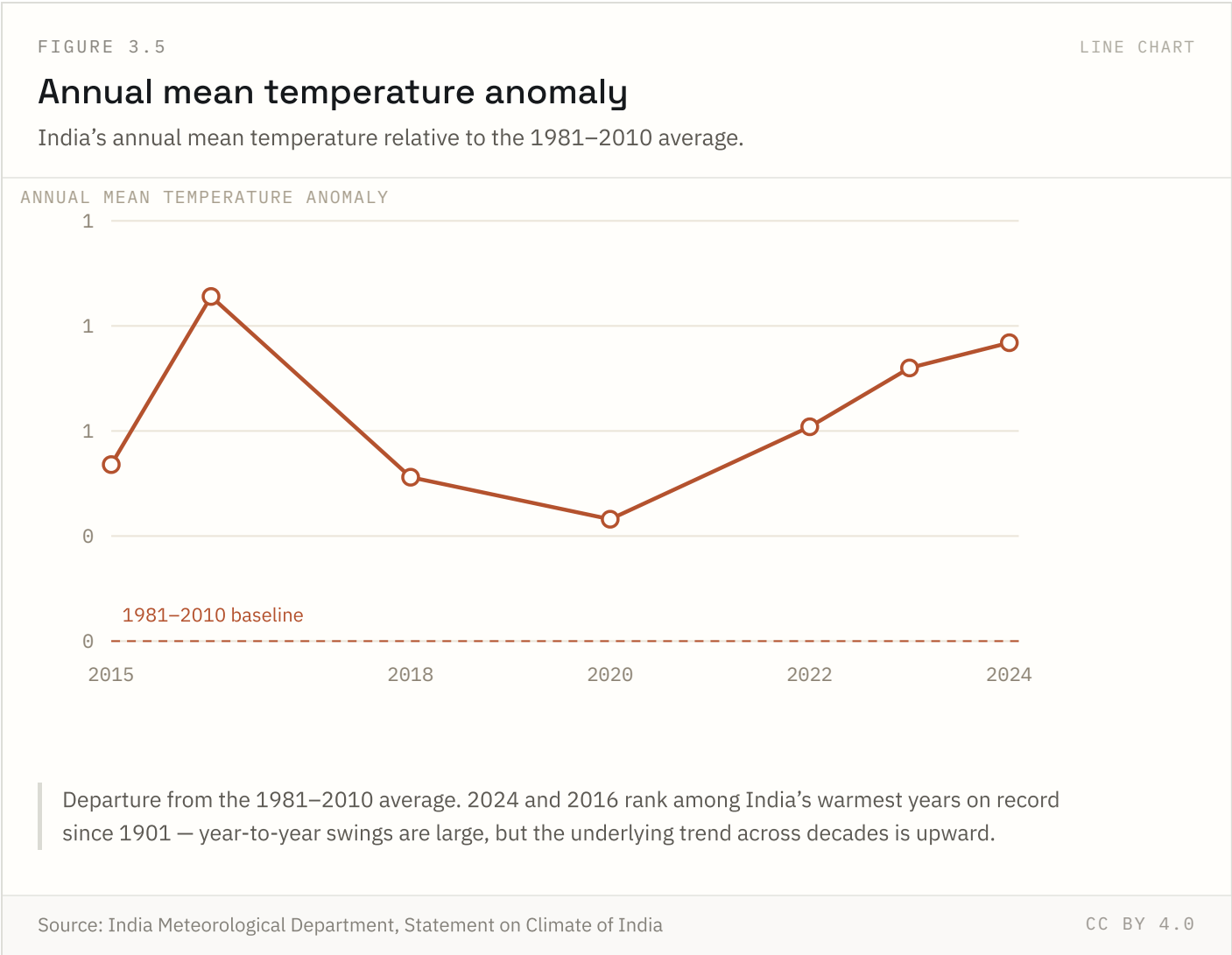


2024 was widely reported as India’s most intense heatwave year in over a decade, with several cities recording temperatures above 45°C. The station-day metric rewards both longer heatwaves and wider geographic spread, so a single very hot, very long event in a densely monitored region can dominate the national total.

Source: India Meteorological Department, Statement on Climate of India Coverage: 2022–2024

Updated: 18 Aug 2026

FIGURE 3.5 NATIONAL ONLY

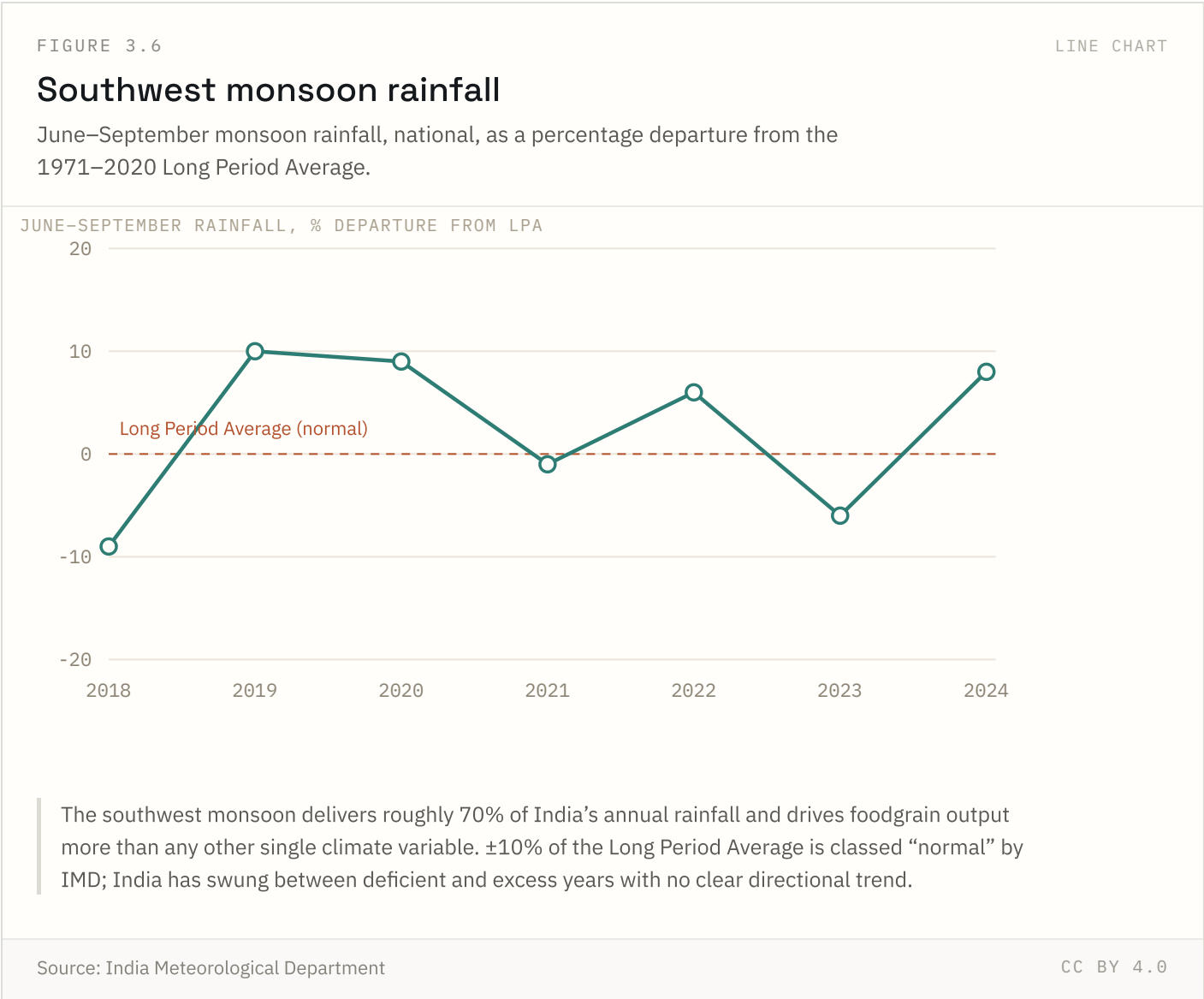


IMD has ranked several recent years among the warmest since national records began in 1901. Annual figures are noisy year to year — a single cool year does not reverse the multi-decade warming trend visible in the underlying data.

Source: India Meteorological Department, Statement on Climate of India Coverage: 2015–2024

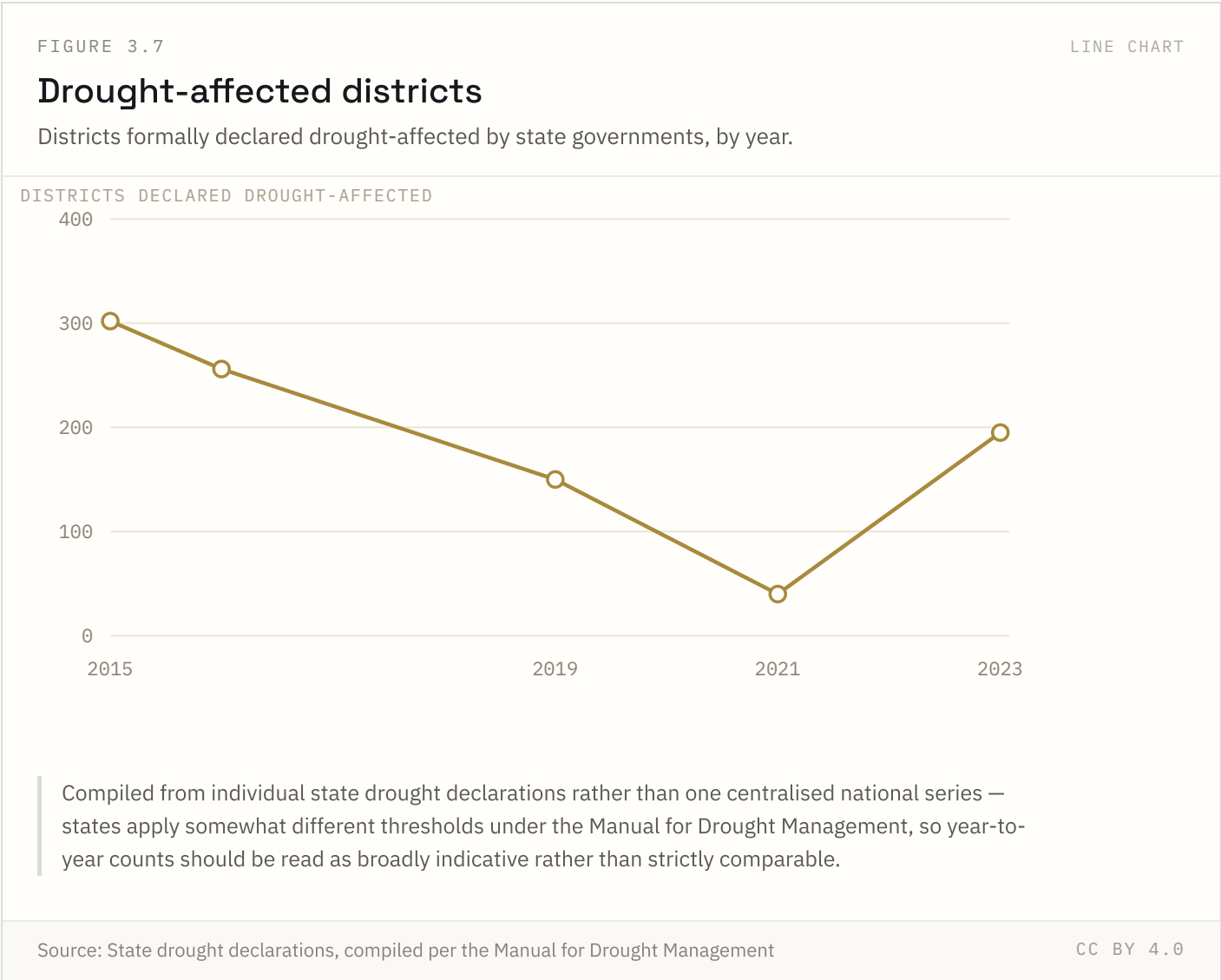
Updated: 18 Aug 2026

FIGURE 3.6 NATIONAL ONLY



IMD classes  $\pm 10\%$  of the LPA as “normal.” The monsoon’s national total can mask sharply uneven regional distribution — a “normal” year on paper can still combine drought in one region with flooding in another.

FIGURE 3.7 NATIONAL ONLY



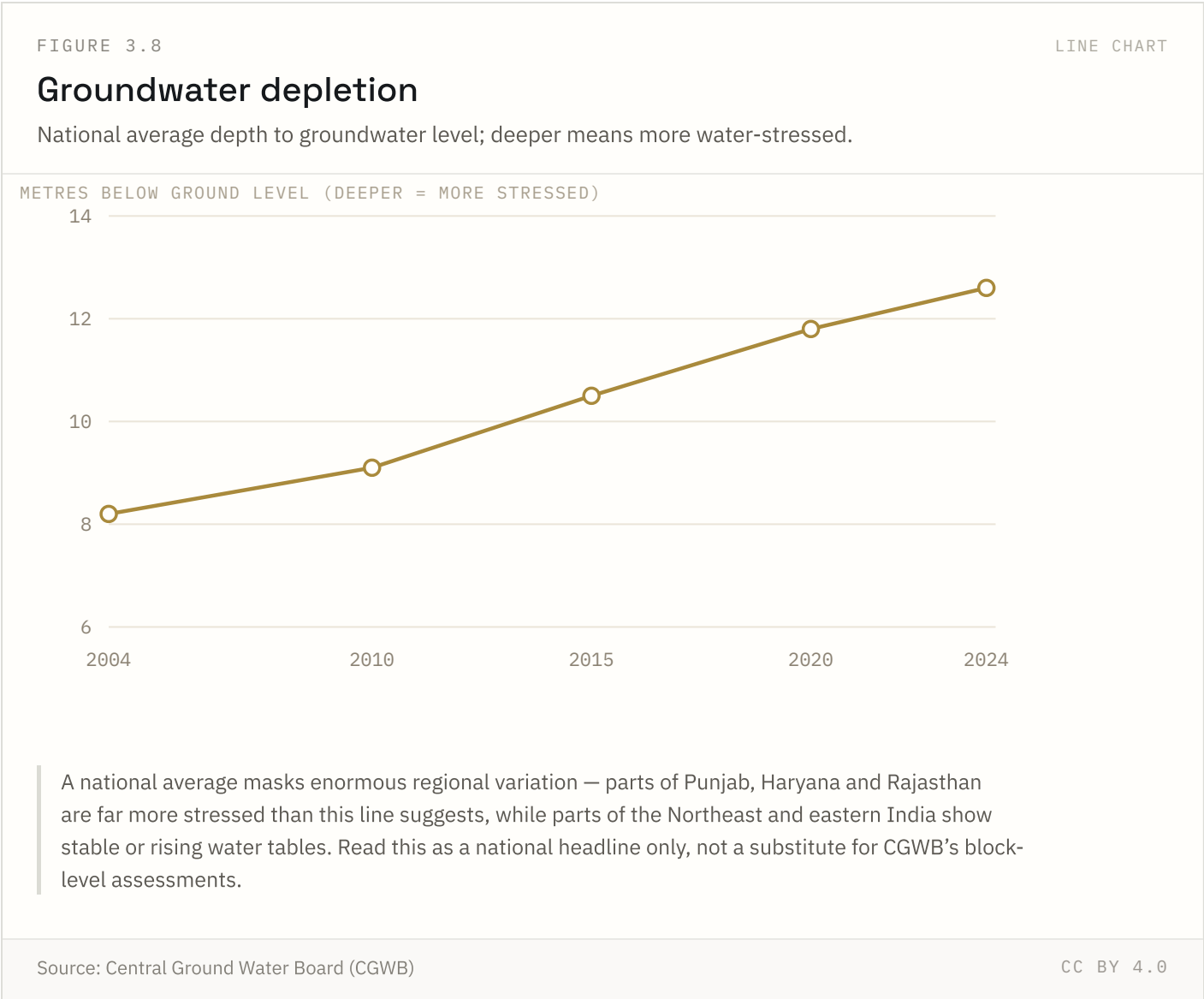
There is no single centralised national drought register — this compiles individual state declarations, which apply the Manual for Drought Management’s criteria with some local discretion. Treat trends as broadly indicative; exact counts vary by source and reporting date.

Source: State drought declarations, compiled per the Manual for Drought Management Coverage: 2015–2023

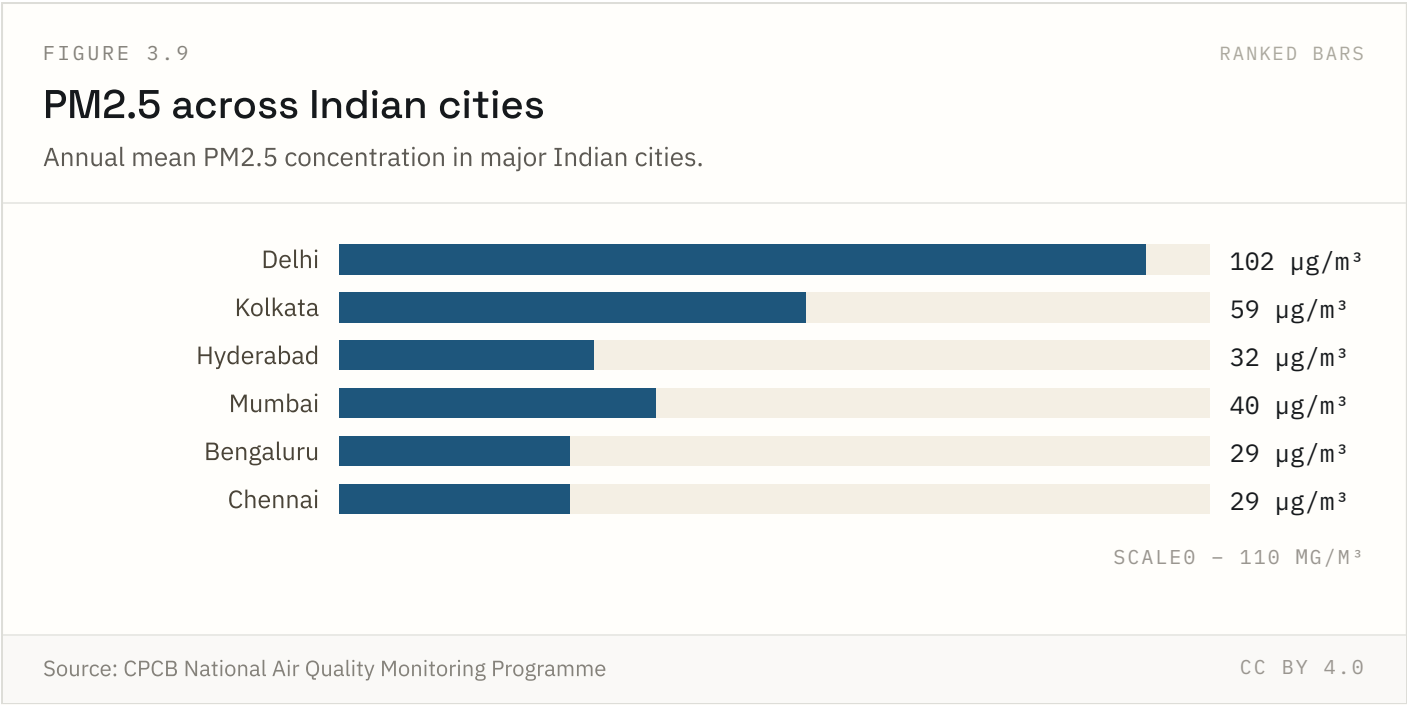
Updated: 18 Aug 2026



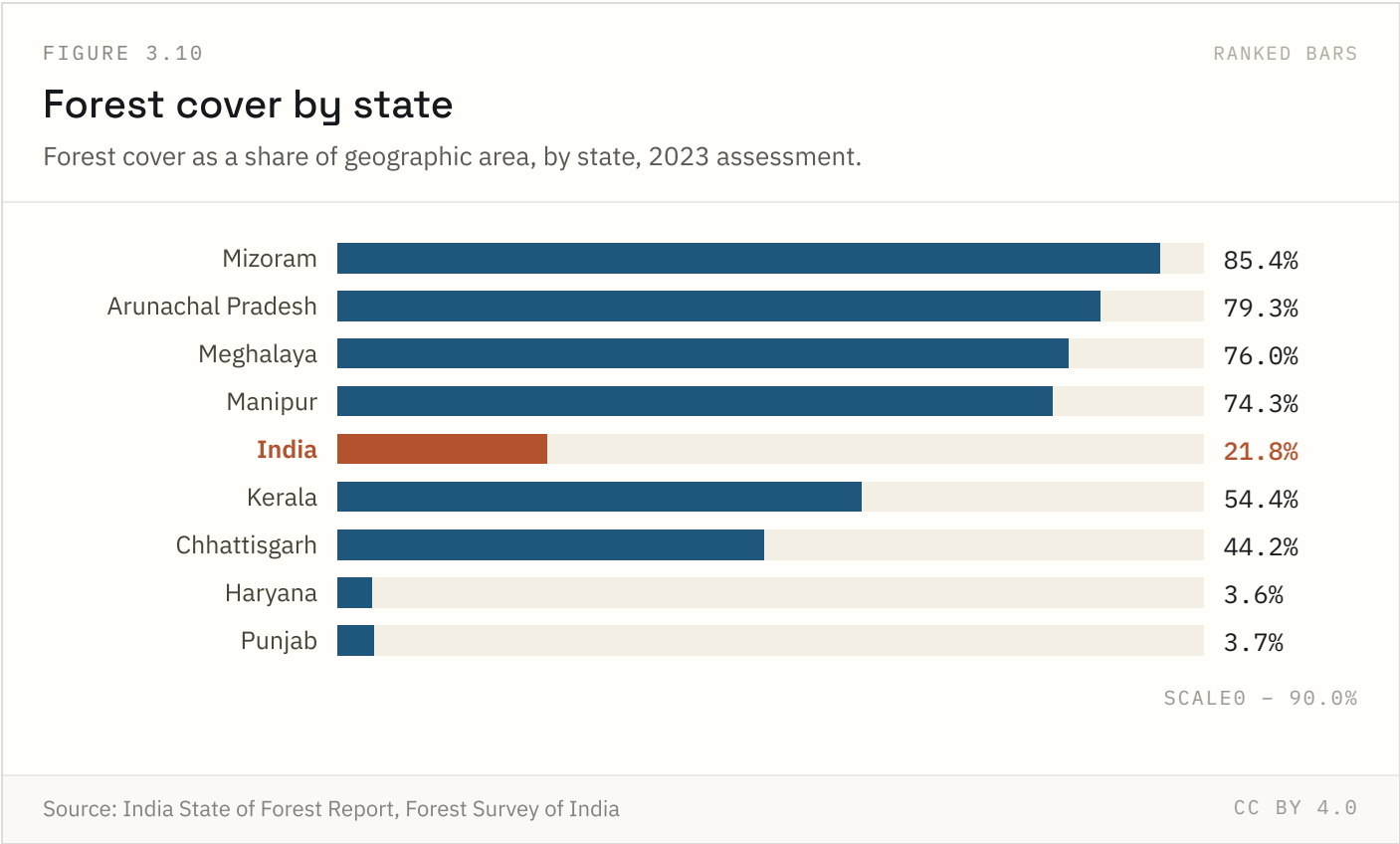
FIGURE 3.8 NATIONAL ONLY



India is the world’s largest extractor of groundwater, used mainly for irrigation. A national average conceals sharp regional divergence — Punjab, Haryana and Rajasthan are critically over-extracted, while parts of eastern India show stable or even rising water tables.



Every city shown exceeds both the WHO guideline (5  $\mu\text{g}/\text{m}^3$ ) and India’s own national standard (40  $\mu\text{g}/\text{m}^3$ ). Delhi’s winter inversion and regional stubble-burning make its annual average an outlier even among India’s other large, heavily polluted cities.



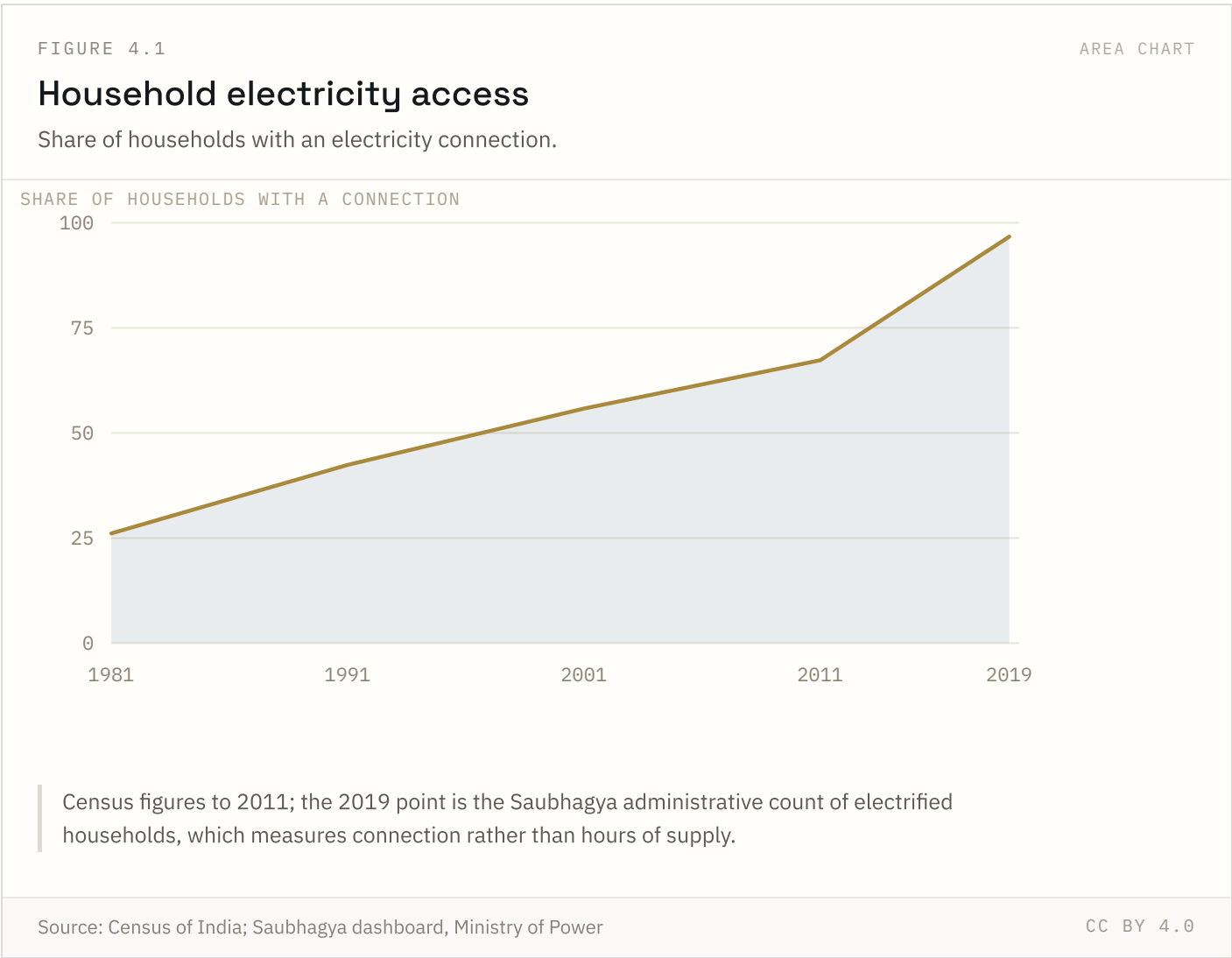
The gap between India’s most and least forested major states spans more than 80 percentage points. Northeastern states carry a hugely disproportionate share of India’s forest cover relative to their land area’s share of the country.

Highest: **Mizoram** at 85.4%. Lowest: **Haryana** at 3.6%.

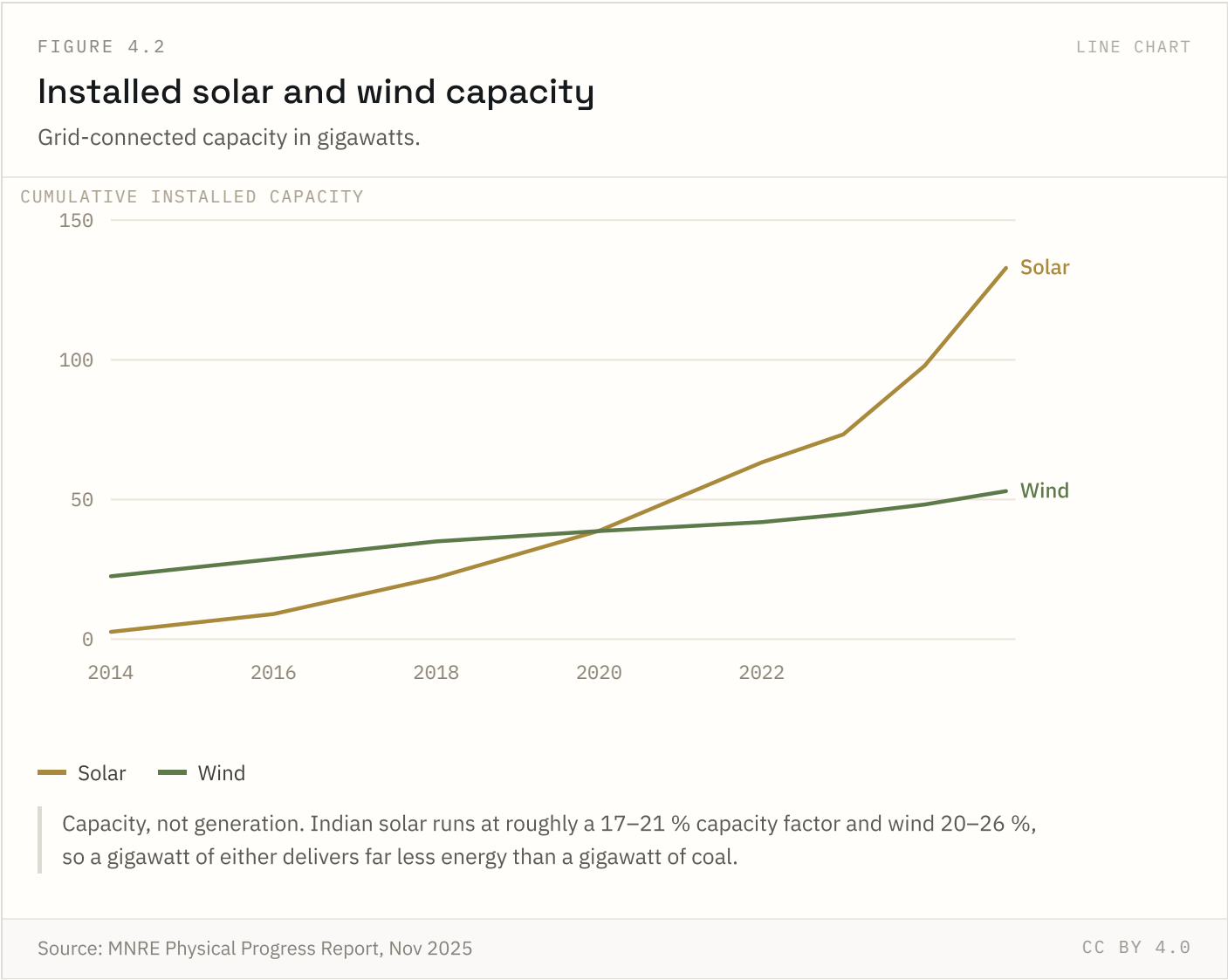
# Energy & electricity

2 of 5 indicators in this chapter carry state-level detail; the rest are published at national level only.

FIGURE 4.1 NATIONAL + STATE

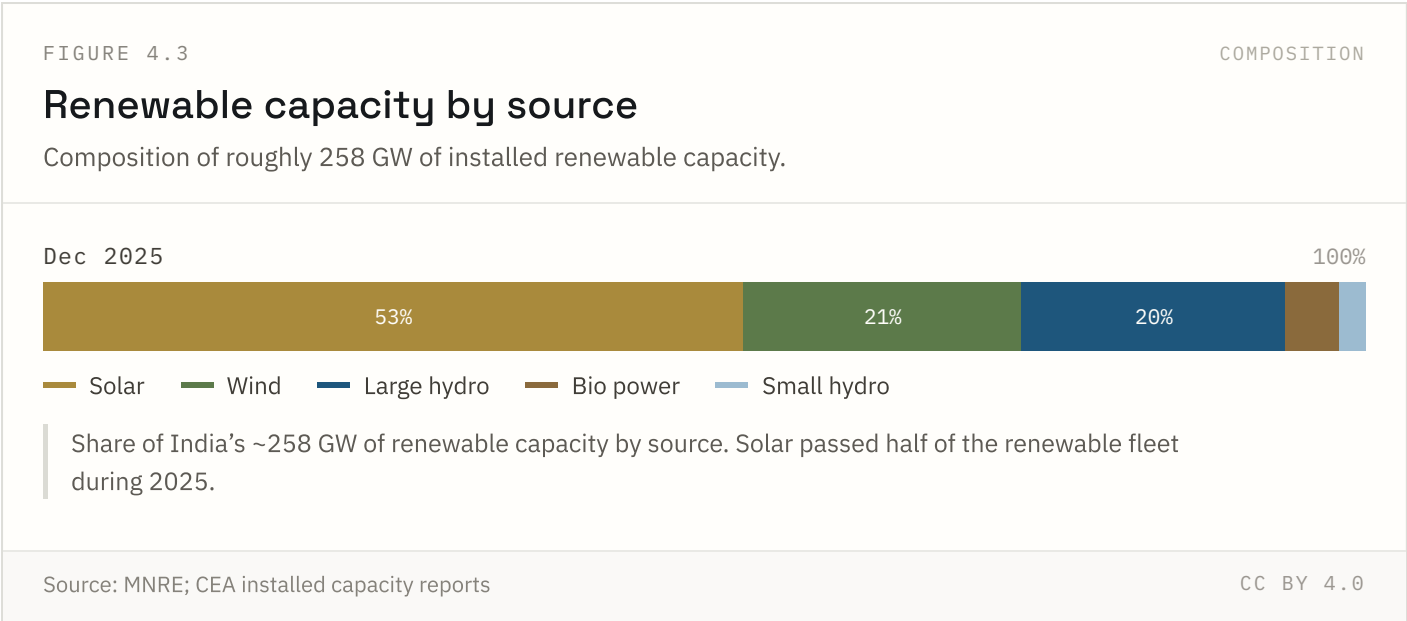


Connection is not the same as supply. Administrative data reports near-universal connection since 2019, but survey data on hours of supply shows large rural shortfalls, especially in Bihar, Jharkhand and Uttar Pradesh.



Capacity is not generation. Solar capacity factors in India run 17–21 per cent, wind 20–26 per cent, so a gigawatt of each delivers far less energy than a gigawatt of coal. Both series are ahead of the trajectory implied by the 2015 targets.

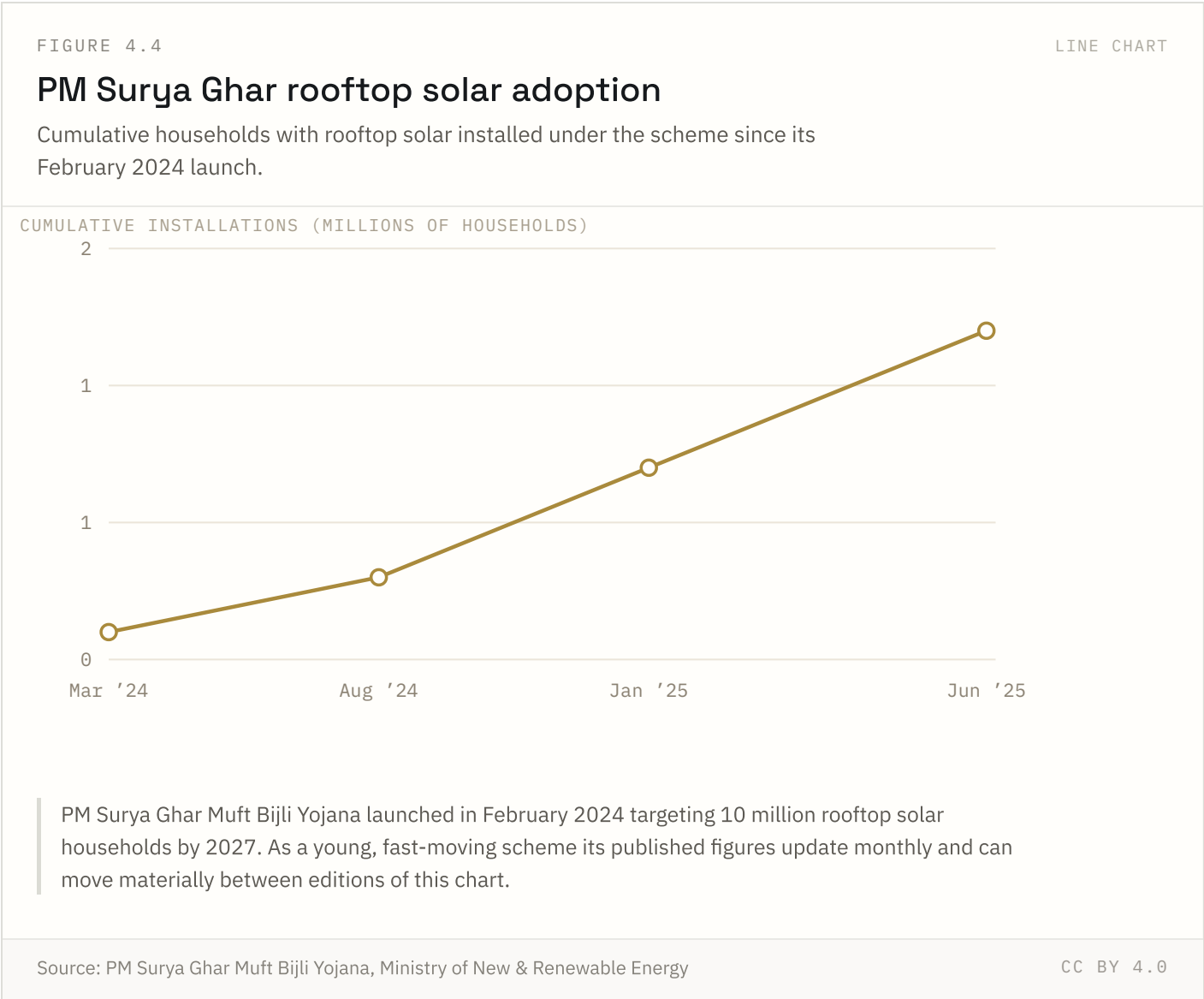
FIGURE 4.3 NATIONAL ONLY



Solar passed half the renewable fleet during 2025. Large hydro, which took seven decades to build, is now smaller than solar, which took eleven years.

Source: MNRE; CEA installed capacity reports Coverage: Dec 2025 Updated: 30 May 2026

FIGURE 4.4 NATIONAL ONLY

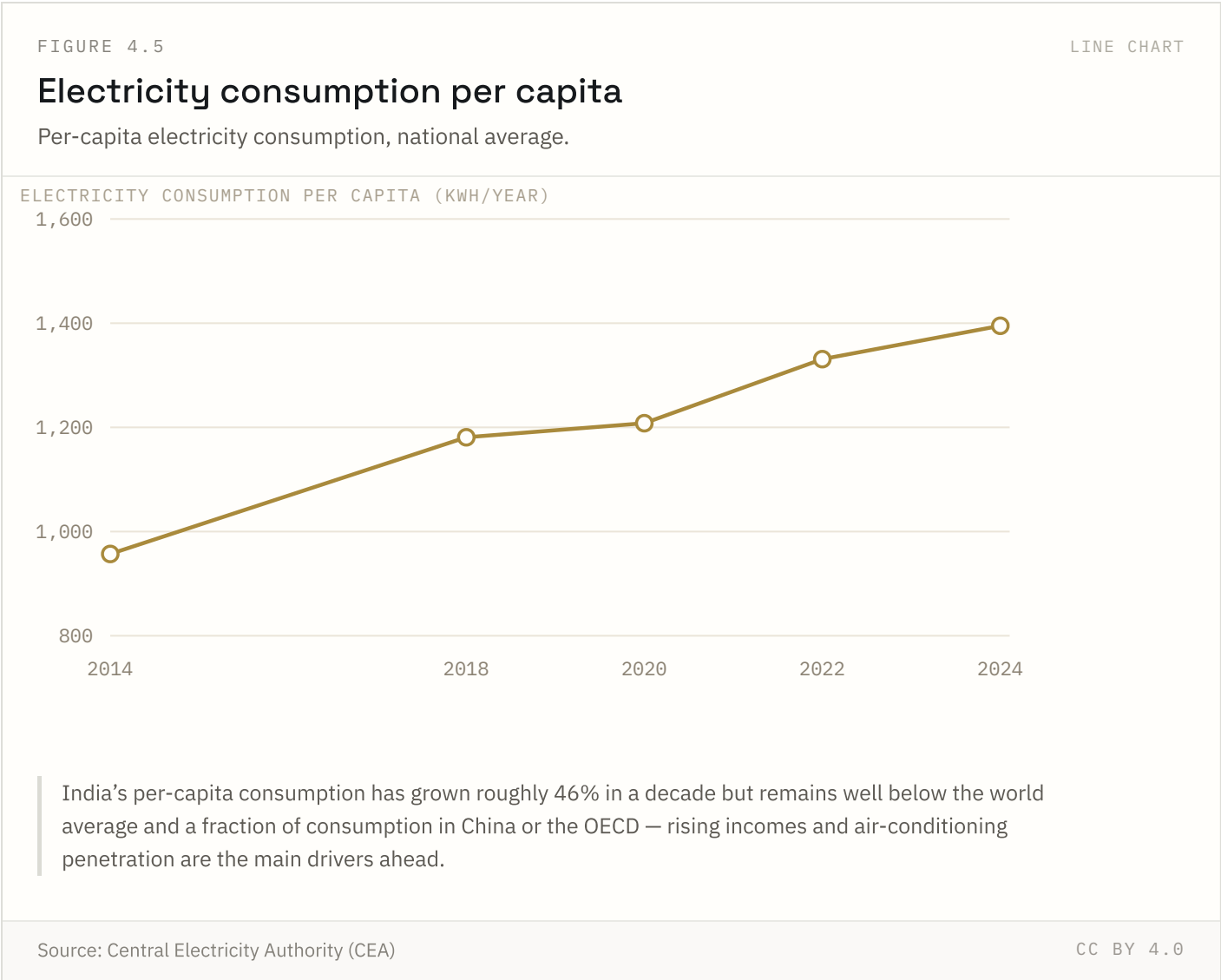


The scheme targets 10 million rooftop-solar households by 2027; adoption so far is a small fraction of that target and figures update frequently as installation data is reconciled with subsidy disbursement.

Source: PM Surya Ghar Muft Bijli Yojana, Ministry of New & Renewable Energy Coverage: Mar 2024–Jun 2025

Updated: 18 Aug 2026

FIGURE 4.5 NATIONAL ONLY



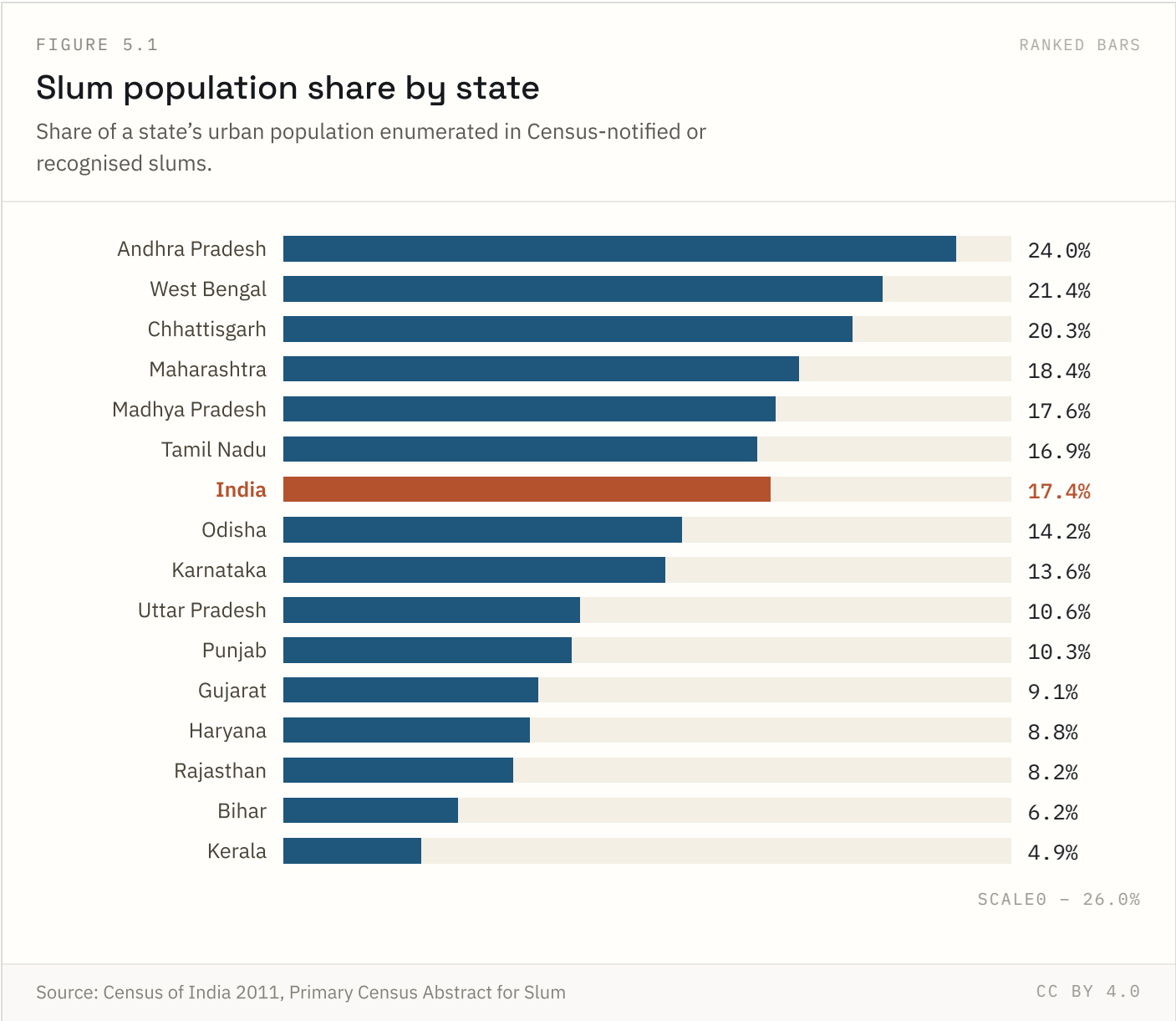
This is a different question from your household electricity access indicator: access asks whether a connection exists; this asks how much power the average person actually uses — a better proxy for living standards and industrial activity than connection rates alone.

Source: Central Electricity Authority (CEA) Coverage: 2014–2024 Updated: 18 Aug 2026



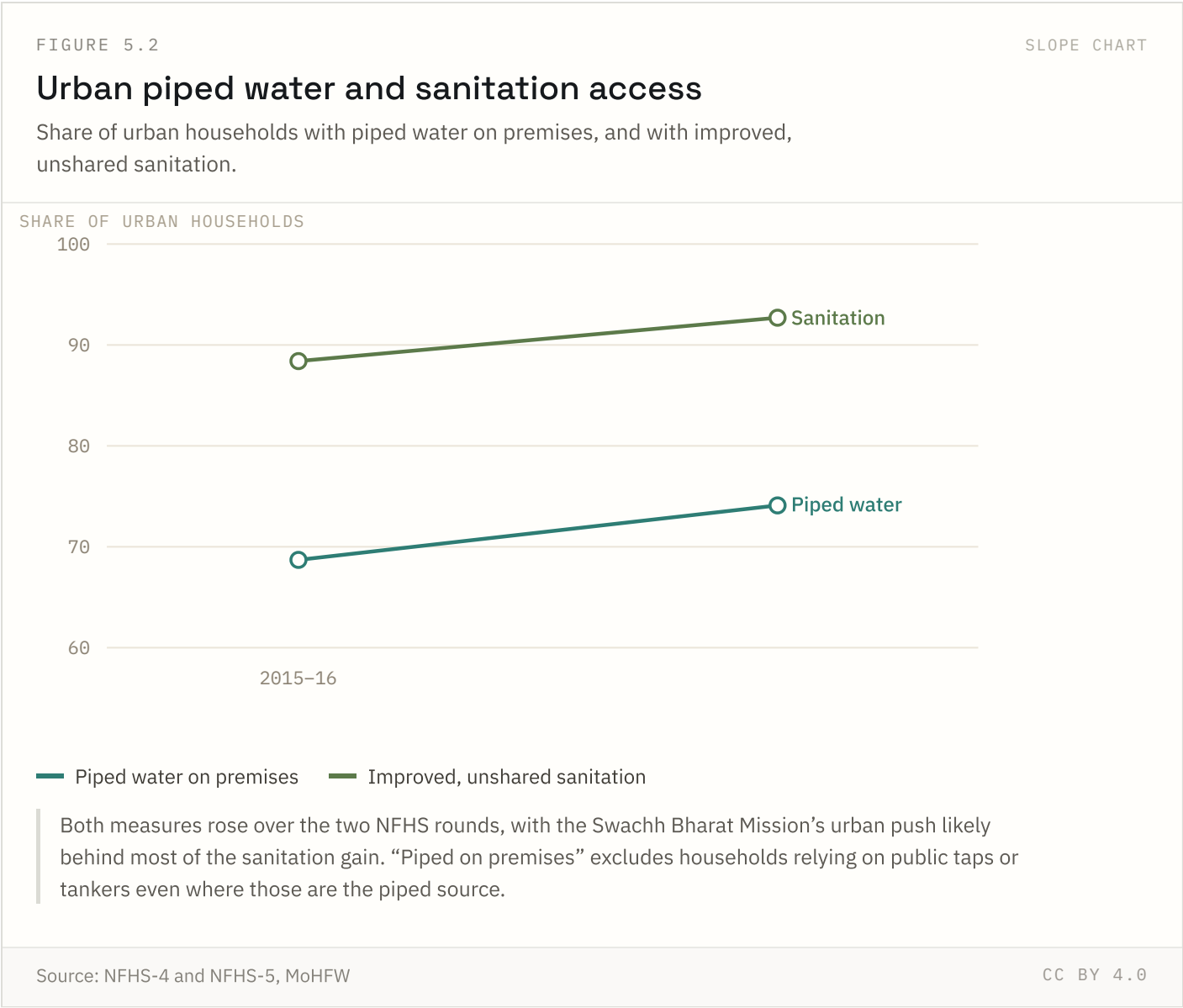
# Urban development & housing

1 of 7 indicators in this chapter carry state-level detail; the rest are published at national level only.

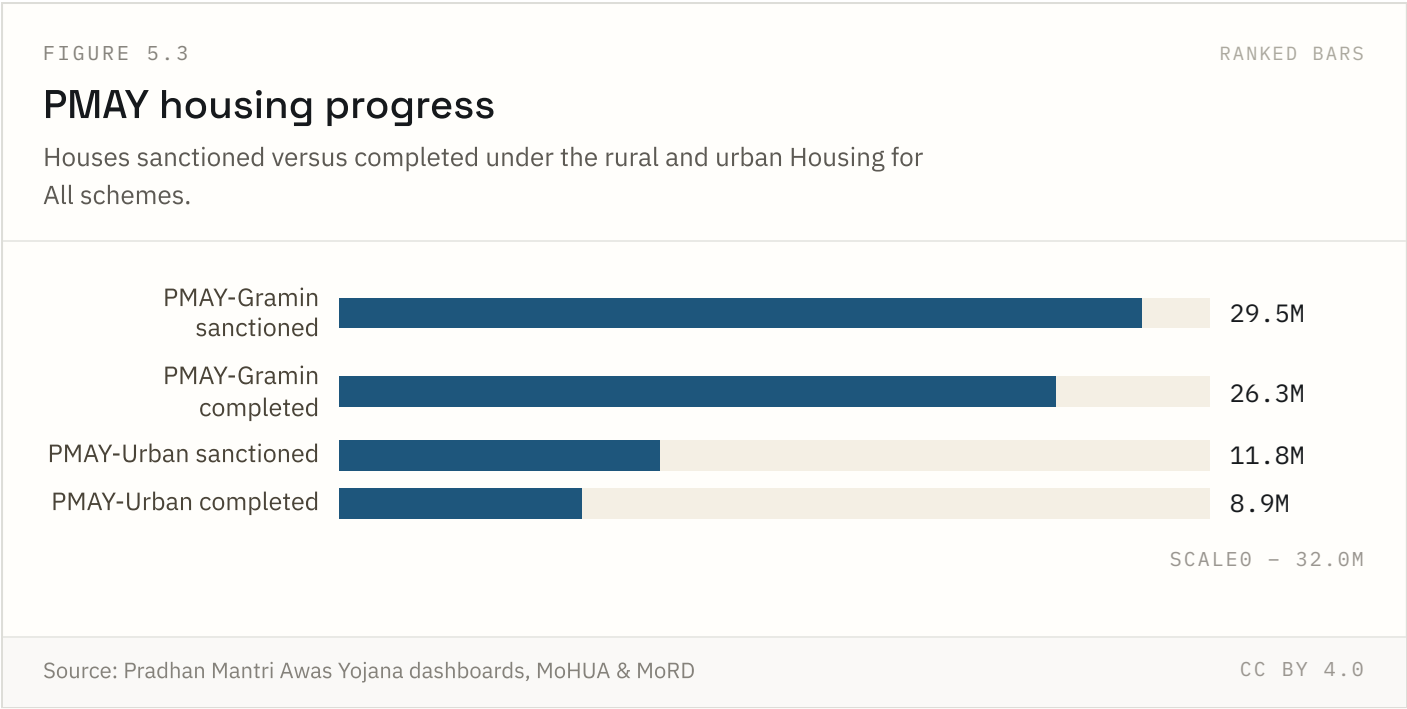


This is the last Census-based slum count available — no Census has been held since 2011, so growth in informal settlements over the past decade is not captured here. Definitions vary by state (notified, recognised and identified slums), which limits strict comparability.

Highest: **Andhra Pradesh** at 24.0%. Lowest: **Kerala** at 4.9%.



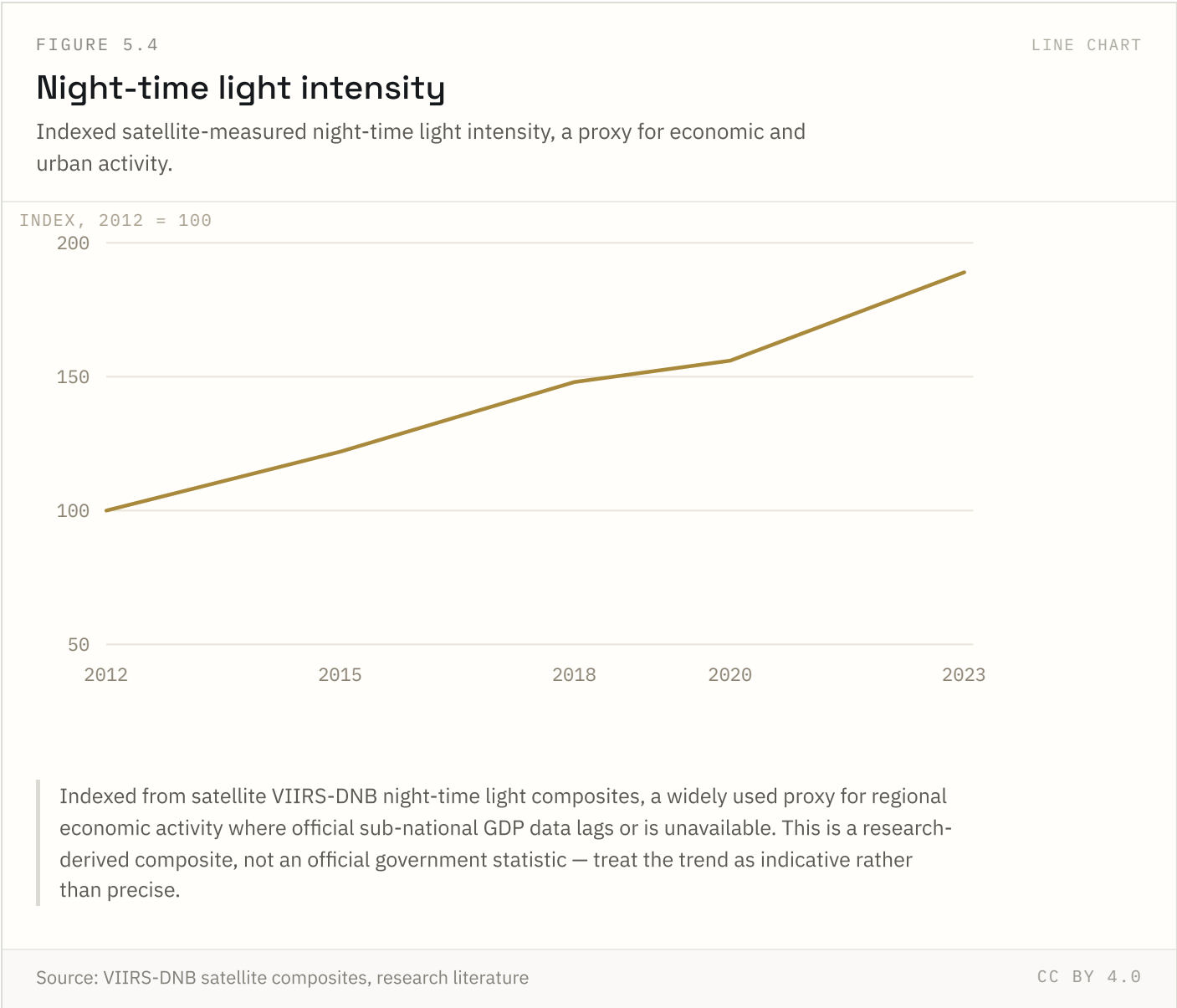
Both series improved between NFHS rounds. “Piped on premises” is stricter than “access to an improved water source” — it excludes households that rely on public taps, tankers or neighbours’ connections even if the underlying source is piped.



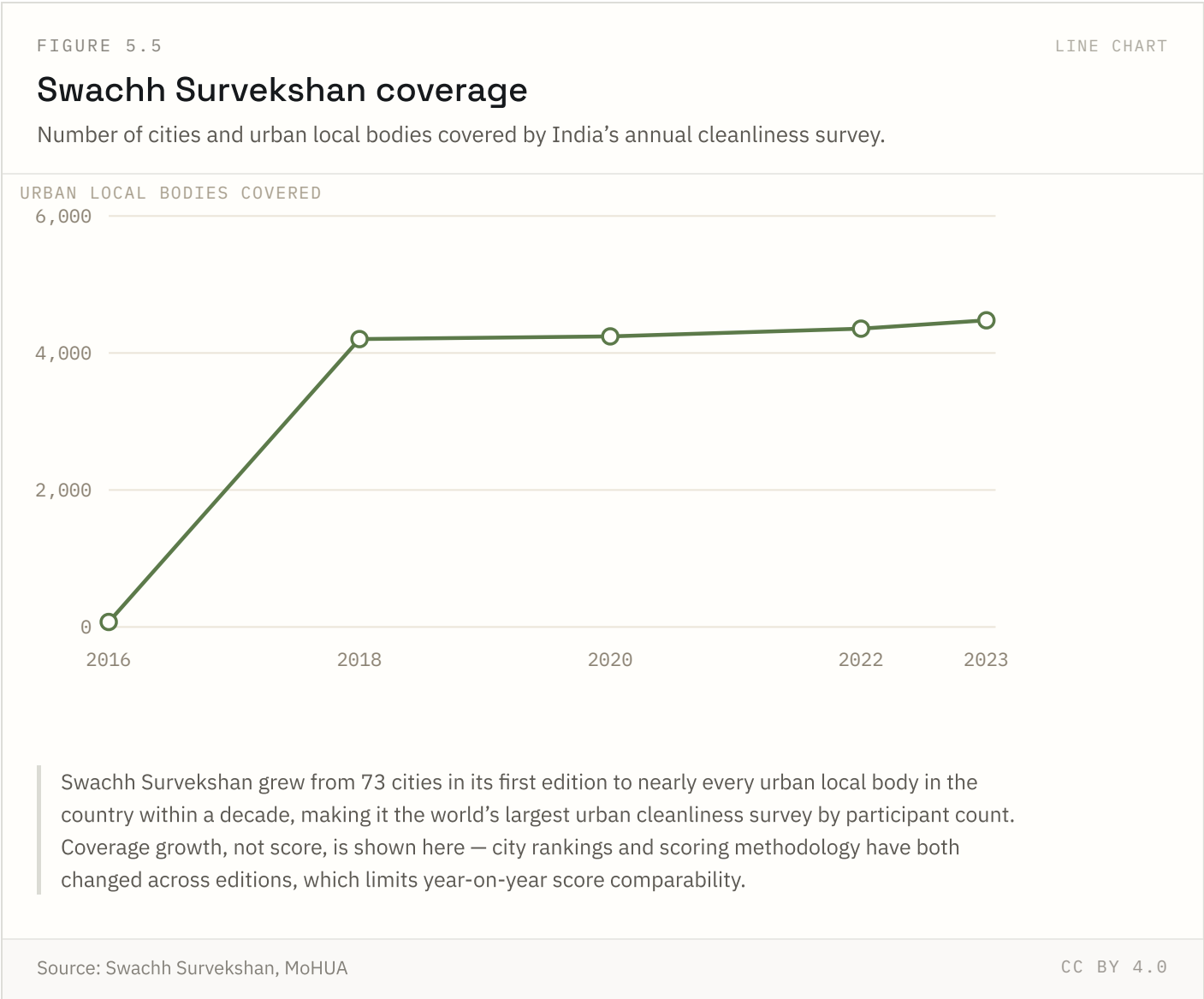
The rural scheme (PMAY-Gramin) is closer to its sanctioned target than the urban scheme (PMAY-Urban), where land acquisition and construction timelines run longer. “Completed” means handed over to the beneficiary, not merely under construction.

Source: Pradhan Mantri Awas Yojana dashboards, MoHUA & MoRD   Coverage: Cumulative to 2025   Updated: 18 Aug 2026

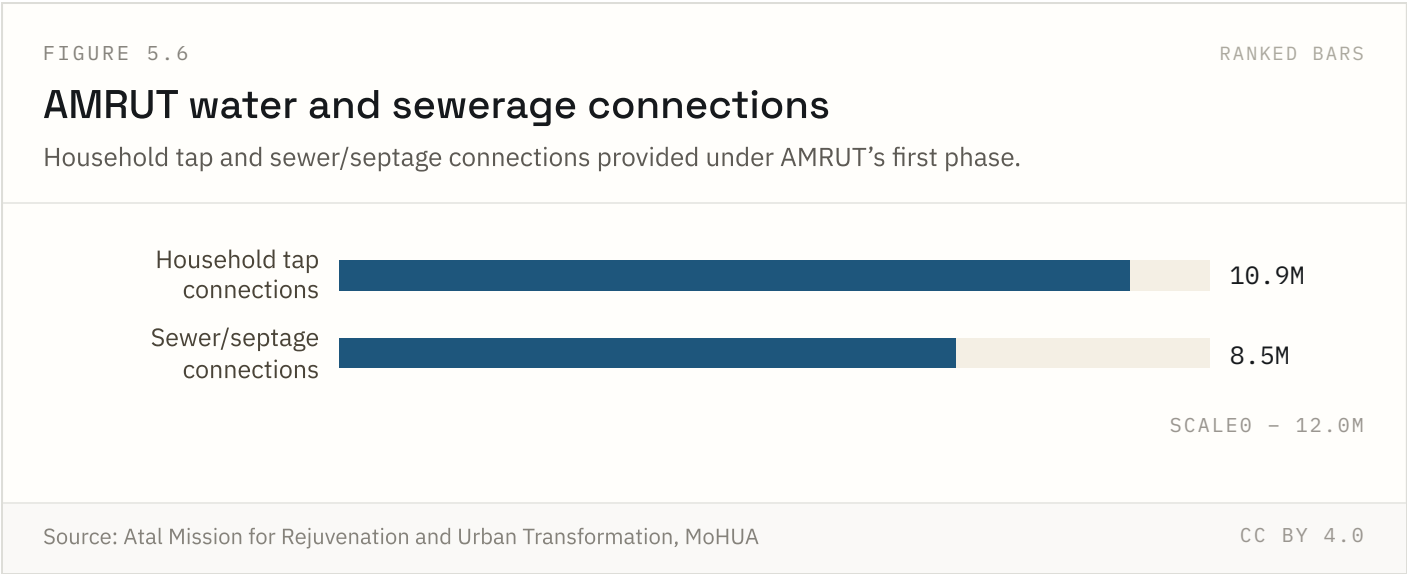
FIGURE 5.4 NATIONAL ONLY



Night-time lights are used by researchers as a fast, sub-national proxy for economic growth where official data is thin or delayed. It captures electrified activity, not output directly, so it can overstate growth in areas that simply gained grid access rather than economic output.



The survey has become close to universal across urban India, which is itself a policy achievement independent of any city’s score. Scoring methodology and weightings have changed across editions, so a city’s rank in different years is not a clean apples-to-apples comparison.

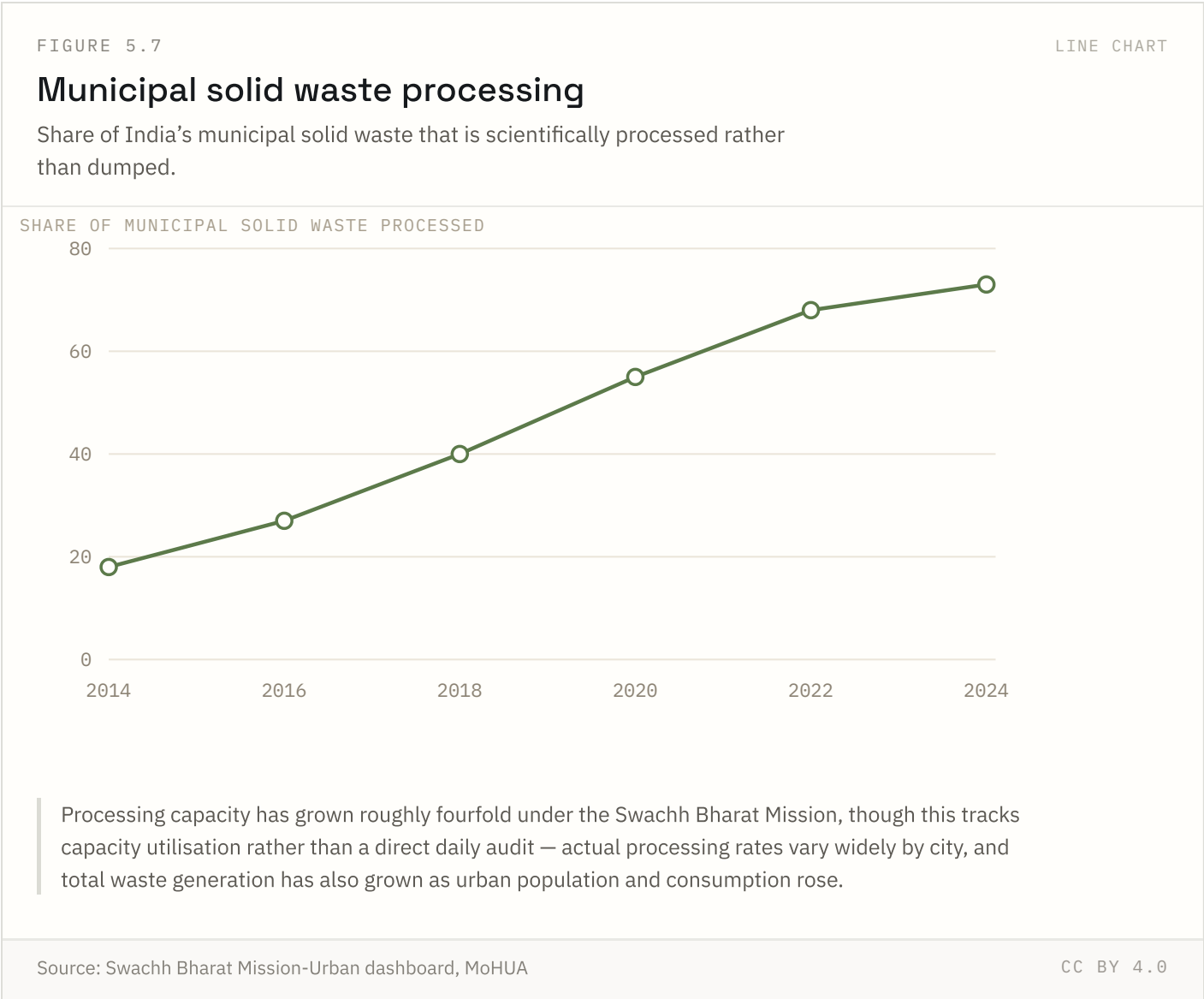


AMRUT 1.0 fell short of its full connection targets by its 2021 close; AMRUT 2.0 (launched 2021) aims at universal coverage. Sewerage lagged water supply throughout the scheme, reflecting the higher cost and longer build time of underground network projects.

Source: Atal Mission for Rejuvenation and Urban Transformation, MoHUA   Coverage: 2015–2021 (AMRUT 1.0)

Updated: 18 Aug 2026

FIGURE 5.7 NATIONAL ONLY



Processing capacity has grown roughly fourfold since 2014, but total waste generation has also risen with urban population and consumption growth — a rising processed share does not necessarily mean a falling absolute quantity dumped in landfills.

Source: Swachh Bharat Mission-Urban dashboard, MoHUA Coverage: 2014–2024 Updated: 18 Aug 2026

## Methodology and limitations

Every figure in this report is reproduced from a published source without adjustment. India Insight Lab does not estimate, model or interpolate the values shown. Each chart carries the source it came from, the years it covers and the date the series was last updated.

Three limitations apply to everything above, and are stated rather than worked around:



- **Geographic depth.** Indicators reach national and, for 9 of 45 series, state level. District and city figures are not included because comparable data is not yet integrated.
- **Comparability across sources.** Series drawn from different sources use different definitions, reference periods and sampling designs. They are shown side by side but are not adjusted onto a common basis.
- **Currency.** Update dates vary by series; the most recent revision date is shown beneath each figure. Some official series lag by several years.

**Nothing in this report is projected.** Where a series ends, the report ends with it. No forecast, scenario or extrapolation has been added to extend a trend beyond its published data.

## References

40 sources used in this report.

|    |                                                               |
|----|---------------------------------------------------------------|
| 1  | Atal Mission for Rejuvenation and Urban Transformation, MoHUA |
| 2  | CPCB National Air Quality Monitoring Programme                |
| 3  | Census of India 2011                                          |
| 4  | Census of India 2011, Primary Census Abstract for Slum        |
| 5  | Census of India 2011; SRS 2021, RGI                           |
| 6  | Census of India, D-series migration tables                    |
| 7  | Census of India; Saubhagya dashboard, Ministry of Power       |
| 8  | Census of India; UN World Population Prospects                |
| 9  | Census of India; UN World Urbanization Prospects              |
| 10 | Central Electricity Authority (CEA)                           |
| 11 | Central Ground Water Board (CGWB)                             |
| 12 | DGCI&S, Ministry of Commerce & Industry                       |
| 13 | DPIIT, FDI Statistics                                         |
| 14 | Global Carbon Budget                                          |
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| 15 | IQAir World Air Quality Report 2024                                        |
| 16 | India Meteorological Department                                            |
| 17 | India Meteorological Department, Statement on Climate of India             |
| 18 | India State of Forest Report, Forest Survey of India                       |
| 19 | MNRE Physical Progress Report, Nov 2025                                    |
| 20 | MNRE; CEA installed capacity reports                                       |
| 21 | MoSPI, Consumer Price Index                                                |
| 22 | MoSPI, National Accounts Statistics (2011–12 series)                       |
| 23 | NFHS-4 and NFHS-5, MoHFW                                                   |
| 24 | National Accounts Statistics, MoSPI                                        |
| 25 | National Population Projections 2011–2036, Registrar General of India      |
| 26 | PM Surya Ghar Muft Bijli Yojana, Ministry of New & Renewable Energy        |
| 27 | Petroleum Planning & Analysis Cell (PPAC)                                  |
| 28 | Pradhan Mantri Awas Yojana dashboards, MoHUA & MoRD                        |
| 29 | RBI Financial Stability Report                                             |
| 30 | RBI Monetary Policy Committee resolutions                                  |
| 31 | RBI Weekly Statistical Supplement                                          |
| 32 | RBI; World Bank India Development Update, April 2026                       |
| 33 | SRS Abridged Life Tables 2017–21, RGI                                      |
| 34 | SRS Statistical Report 2021, RGI                                           |
| 35 | State drought declarations, compiled per the Manual for Drought Management |
| 36 | Swachh Bharat Mission-Urban dashboard, MoHUA                               |
| 37 | Swachh Survekshan, MoHUA                                                   |
| 38 | Trade Intelligence and Analytics portal, Department of Commerce            |
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| 39 | VIIRS-DNB satellite composites, research literature |
| 40 | World Bank India Development Update, April 2026     |

Source URLs, licence terms and retrieval dates are not yet recorded for every series. Adding them is scheduled work, and until it is done these references name the publication rather than link to it.